Physics-Infused Reduced-Order Modeling for Analysis of Multi-Layered Hypersonic Thermal Protection Systems With Imperfect Thermal Contacts

Abstract

This work presents a systematic design and implementation of the physics-infused reduced-order modeling (PIROM) framework for optimal inference and reduced-order modeling of hypersonic TPS with unknown thermal contact resistances (TCRs). The PIROM framework is designed to leverage the strengths of both physics-based and data-driven modeling paradigms, by infusing physics-based models with data-driven components to enhance their predictive capabilities and generalizability. Inference heat conduction problems (IHCPs) are solved using the PIROM framework to infer TCRs from internal temperature measurements, while providing accurate predictions of transient TPS temperature dynamics under varying boundary conditions and material properties. Particularly, experimental limitations are considered in the design of the PIROM framework, including noisy observables and thermocouple placement limitations. The performance of the PIROM framework is benchmarked against traditional physics-based modeling approaches (e.g, one-dimensional FEM) and data-driven surrogate models (e.g., neural ordinary differential equations) for both inferring contact conductances and transient TPS temperature dynamics.

1 Introduction

What is the importance of hypersonic TPS? Hypersonic thermal protection systems (TPS) play a critical role in safeguarding hypersonic vehicles as they are exposed to extreme aerodynamic heating during flight. Therefore, accurate modeling and prediction of TPS behavior become essential for designing effective thermal protection strategies. Nonetheless, the highly non-linear fluid-thermal-structural interactions, temperature-dependent material property variations, and uncertainties in thermal contact resistances (TCR) at the interfaces of the TPS layers, pose significant challenges for traditional modeling approaches such as finite-element methods (FEM), as well as reduced-order (ROM) and surrogate (SM) modeling techniques.

What is the new problem that we want to solve? Particularly, a critical artifact in transient reduced-order modeling of hypersonic TPS is the presence of TCRs. Due to manufacturing limitations, when two solids are brought together to form an interface, the true solid-to-solid contact area between them is generally a small fraction of the apparent area over which they meet [Minges1970]. The net effect of such imperfect contacts is a temperature discontinuity at the interface that is in general challenging to predict, and that significantly affects the heat transfer across the interface. The identification of TPS properties from internal temperature measurements belongs to a class of *inverse heat conduction problem* (IHCP) [Deng2026], which are known to be ill-posed and sensitive to measurement noise, and thus require careful regularization and robust inference techniques for accurate estimation of TCRs. The challenges posed by IHCPs can introduce significant uncertainties in transient temperature predictions, and thus compromise the reliability of ROMs for design, analysis, and optimization of hypersonic TPS [Fang2023]. Therefore, there is a critical need for *efficient*, *accurate*, and *generalizable* modeling approaches of TPS that can infer TCRs, while providing reliable predictions of internal TPS transient temperaures under varying boundary conditions and material properties.

What has been done in the literature for inference and simulation? Attempts to model hypersonic TPS with varying TCRs have taken various forms in the literature, ranging from traditional deterministic physics-based and probabilistic approaches

For instance, in Ref. [Fang2023], a Lattice Boltzmann Method (LBM) is used to model the transient heat transfer and solid-liquid phase change in a hypersonic TPS, where the TCRs are assumed known, and are shown to significantly delay the melting points of phase changing materials. In Ref. [Zhang2024], authors take one step further by treating the TCRs as unknown parameters, and a finite-element thermo-mechanical model of a three-dimensional reusable heat-pipe cooled thermal protection system is developed to perform Levenberg-Marquardt gradient-based inference of internal TCRs using boundary temperature measurements. A similar approach is adopted in Ref. [Brociek2024], where a finite-difference one-dimensional model of a three-layered TPS is developed to solve the forward problem, and a Levenberg-Marquardt for a Tikhonov regularized optimization is used to solve the inverse problem of inferring TCRs from internal temperature measurements.

Probabilistic approaches to solving the inverse problem of inferring TCRs have also been proposed in the literature. For instance, in Ref. [Gao2026], a Bayesian Network (BN) with sequential importance resample (SIR) is leveraged for proabilistic inversion of TCRs, where the BN is implemented by leveraging a finite-element model of the multi-layered TPS for forward simulations. A BN is also leveraged in Ref. [Deng2026] to infer TCRs, where a neural network is used as a surrofate model to predict internal temperature dynamics due to the computational cost of the FEM-based forward model.

The use of scientific machine learning (SciML) techniques for modeling hypersonic TPS with unknown TCRs has limited exploration in the literature.

.... Other X approaches such as those in Refs.[X] ...

While these methods offer a high-fidelity representation of the underlying physics, highly refined FEMs result in large-scale systems that are computationally expensive to solve, and thus are not suitable for design, analysis, and optimization of hypersonic TPS.

Other data-driven approaches such as those in Refs.[X] ..., leverage Bayesian Networks (BNs) to learn mappings from boundary conditions and thermocop

machine learning techniques to learn the underlying dynamics of the TPS system directly from data, while inferring the TCRs. For instance, in Refs. [X]...

While these methods offer computational efficiency, they often require large amounts of data for training, and may not generalize well to unseen conditions, such as varying boundary conditions and material properties. Moreover, they may not provide insights into the underlying physics of the system, which can be critical for design and optimization.

What is the present work doing? This work presents a systematic design, implementation, and benchmarking of the PIRM for modeling hypersonic TPS with unknown TCRs under experimental constraints such as noisy observables and thermocouple placement limitations.

Objectives of the work The objectives of this work are to,

1. Demonstrate the systematic application of PIROM for modeling hypersonic TPS with unknown contact resistances, emphasizing on the simultaneous inference of material properties and hidden dynamics with limited data.

2. Benchmark the performance of PIROM against traditional physics-based modeling approaches (e.g, one-dimensional FEM) and data-driven surrogate models (e.g., neural ordinary differential equations) for both inferring contact conductances and transient TPS temperature dynamics.

3. Provide insights into the optimal design of physics-based models for inference of contact conductances, including the resolution of the physics-based model, noise levels, and constraints with the placement of thermocouples.

2 Modeling of Hypersonic Thermal Protection Systems

The following discussion presents the modeling of the transient heat conduction problem in multi-layered hypersonic thermal protection systems (TPS) with imperfect interfaces applications. Specifically, three different approaches of decreasing fidelity are presented for solving the governing equations: (i) a high-resolution three-dimensional FEM termed the *full-order model* (FOM), (ii) a coarse one-dimensional FEM model with linear elements termed the *one-dimensional FEM* (1D-FEM), and (iii) a finite-volume-based lumped capacitance model (LCM) with interfacial temperature jump dynamics termed the *reduced-physics model* (RPM). The three solution strategies are described in detail next.

2.1 Governing Equations

Consider the generic domain $\Omega \subset \mathbb{R}^d$ show in Fig. 1. A heat flux q_b (i.e., Neumann boundary condition) is prescribed on boundary e_q, while a temperature T_b (i.e., Dirichlet boundary condition) is prescribed on boundary e_T, and $\partial\Omega = \cup_{i=1}^{N} \partial\Omega_i$ where $e_{iq}, e_{iT} \subset \partial\Omega_i$. The transient heat transfer is governed by the following energy PDE,

$$\rho c_p \partial_t T - \nabla \cdot (\mathbf{k}(T)\nabla T) = 0,\ x \in \Omega \tag{1a}$$

$$-\mathbf{k}(T)\nabla T \cdot \mathbf{n} = q_b(r,t),\ r \in e_q \tag{1b}$$

$$T(r,t) = T_b(r,t),\ r \in e_T \tag{1c}$$

$$T(r,0) = T_0(r),\ x \in \Omega \tag{1d}$$

where the density ρ is constant, while the heat capacity c_p and thermal conductivity $\mathbf{k} \in \mathbb{R}^{d\times d}$ are temperature-dependent. The TPS domain is further divided into N non-overlapping components $\{\Omega_i\}_{i=1}^{N}$, with N_Γ different imperfect interfaces between components. Specifically, to facilitate subsequent model derivations, each domain Ω_i is further subdivided into: (i) a *contact subdomain* Ω_i^c consisting of the portion of TPS layer that is in direct contact with a neighboring layer across and imperfect interface, and (ii) a *bulk subdomain* Ω_i^b consisting of the remaining interior bulk material; clearly, $\Omega_i = \Omega_i^b \cup \Omega_i^c$, and $\Omega_i^b \cap \Omega_i^c = \emptyset$.

For example, Fig. 1 contains $N_C = 3$ components and $N_\Gamma = 3$ imperfect contacts between components. Such partition of the TPS is introduced to systematically model the *bulk* and *contact* heat transfer as separate but coupled mechanisms, which becomes critical in subsequent sections to infer TCRs and learn the hidden dynamics in the PIROM. The following discussion introduces three different approaches of decreasing fidelity to modeling the hypersonic TPS system: (i) a

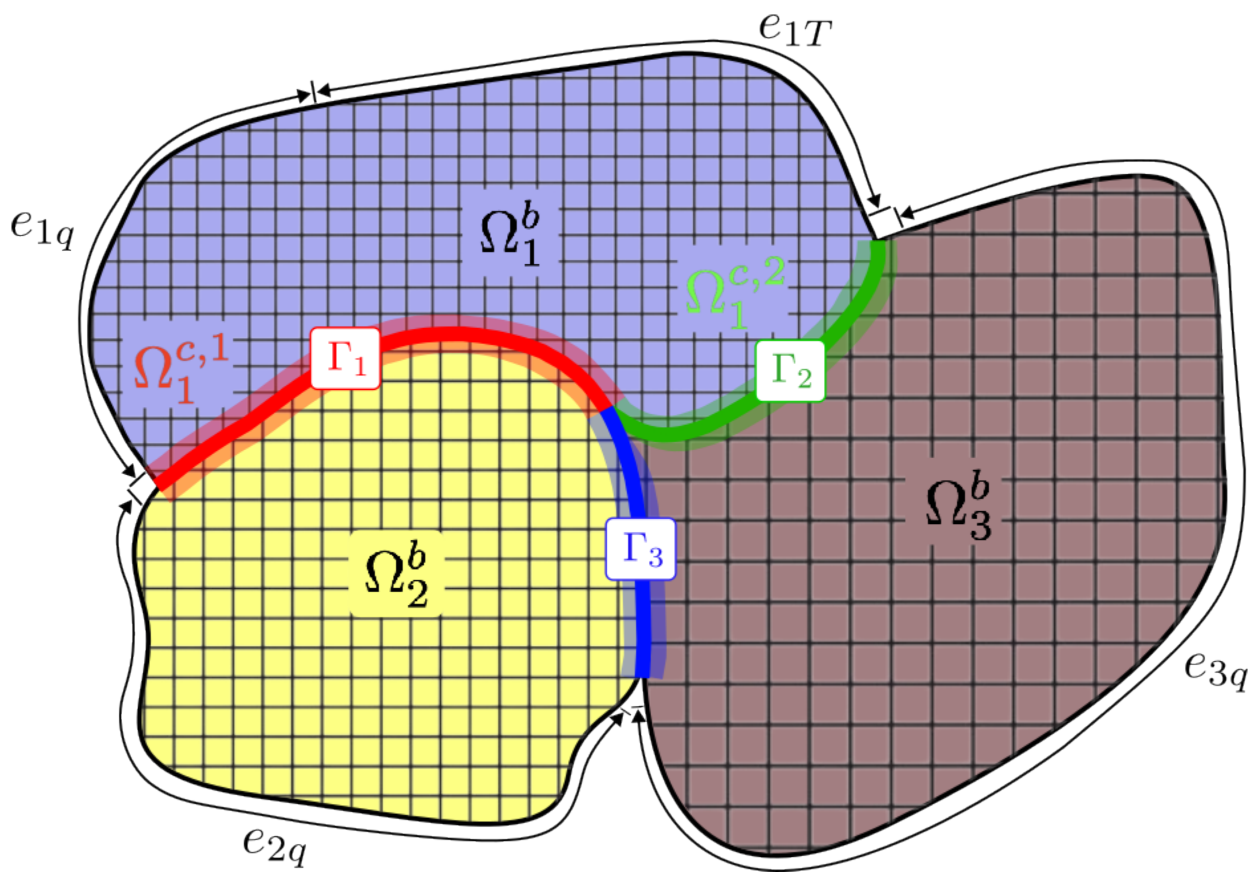


Figure 1: Schematic of the general TPS domain Ω composed of $N_C = 3$ components and $N_\Gamma = 3$ imperfect contact interfaces between components.

FOM based on a three-dimensional FEM, (ii) a coarse one-dimensional FEM, and (iii) a coarse finite-volume, i.e., lumped capacitance model (LCM).

2.2 Full-Order Finite-Element Model

The FOM is obtained by spatially discretizing the governing PDE using a Discontinuous Galerkin (DG) FEM, to result in a high-dimensional system of ODEs that model the transient temperature dynamics of the TPS. The DG-FEM formulation is leveraged for discretization as it: (i) provides a systematic framework for modeling multi-layered TPS with complex geometries, varying material properties, and interfacial TCRs, and (ii) is amenable to systematic model-order reduction via coarse-graining. Note that the high-fidelity numerical simulations in Sec. [X] are performed using standard FEM instead, and the equivalence between DG and standard FEM is noted upon their convergence.

Consider a conforming mesh partition of the TPS domain Ω, were each element belongs to one and only one of the components $\{\Omega_i\}_{i=1}^N$, and where each domain Ω_i is further divided into *bulk* and *contact* sub-domains, i.e., $\Omega_i = \Omega_i^b \cup \Omega_i^c$. Denote the collection of elements on the bulk sub-domains as $\{B_i\}_{i=1}^B$, and on the contact sub-domains as $\{W_i\}_{i=1}^K$, so that $\{E_i\}_{i=1}^M$ with $M = K + B$ corresponds to the complete collection of elements over the entire TPS. Moreover, the perfect, imperfect, Neumann, and Dirichlet boundary interfaces are denoted as e_{ij}, Γ_{ij}, e_{iq}, and e_{iT}, respectively. Lastly, for an element E_i, $|\partial E_i|$ denotes the length ($d = 2$) or area ($d = 3$) of a boundary ∂E_i.

On the i-th element, use a set of P basis functions, such as polynomials, to represent the temperature field as,

$$T_i(r,t) = \sum_{p=1}^{P} \boldsymbol{\phi}_p(r) x_{i,p}(t), \qquad r \in E_i \tag{2}$$

where $\{x_{i,p}\}_{p=1}^{P}$ corresponds to the states of element E_i. Without loss of generality, let the basis functions on element E_i satisfy the orthogonality condition $\int_{E_i} \phi_p^i \phi_q^i d\Omega = |E_i|\, \delta_{pq}, p \neq q$, where δ_{pq} is the Kronecker delta. Furthermore, to aid in subsequent model-order reduction derivations, choose $\phi_1^i = 1$; therefore, by orthogonality, $\int_{E_i} \phi_p^i d\Omega = 0$ for $p > 1$. The first basis function therefore captures the volume-averaged temperature of the element, while the remaining basis functions capture the spatially varying temperature distributions within the element.

Using standard variational procedures, e.g., [Pernet2018] as detailed in Appendix [X], the full governing equation is denoted as,

$$\boldsymbol{\mathcal{A}}(\mathbf{x})\dot{\mathbf{x}} = \boldsymbol{\mathcal{B}}\left(\mathbf{x}\right)\mathbf{x} + \boldsymbol{\mathcal{F}}(t) \tag{3}$$

where $\mathbf{x} = \left[\mathbf{u}, \mathbf{w}\right]^\top \in \mathbb{R}^{MP}$ with $M = B + K$ includes the states for B bulk and K contact elements, i.e., $\boldsymbol{\mathcal{A}} \in \mathbb{R}^{MP\times MP}$ and $\boldsymbol{\mathcal{B}} \in \mathbb{R}^{MP\times MP}$ are the temperature-dependent mass and stiffness matrices, and $\boldsymbol{\mathcal{F}}(t)$ corresponds to the contributions from the boundary conditions, respectively.

2.3 One-Dimensional Finite-Element Model

Next, a *one-dimensional* FEM (1D-FEM) is introduced as a low-order representation of the transient thermal dynamics of the TPS. The model employs a coarse spatial discretization in the through-thickness direction, where each TPS layer is represented using a single first-order element. Such one-dimensional approximations are commonly adopted for TPS analysis because the dominant temperature gradients generally develop along the thickness direction, while the in-plane gradients are comparatively small [Gao2026].

Consider a TPS composed of N non-overlapping layers $\{\Omega_i\}_{i=1}^{N}$, where within each layer, the temperature distribution is approximated as in Eq. (2) with $P = 2$. For instance, in the i-th element, $\boldsymbol{\phi}_i(x) = \left[\phi_{i,1}(x), \phi_{i,2}(x)\right]^\top$ contains linear trial functions, and $\boldsymbol{u}_i(t) = \left[u_i^-(t), u_i^+(t)\right]^\top$ contains the corresponding left and right nodal temperature states. Collecting the states for all layers into a single vector yields $\boldsymbol{u} = \left[\boldsymbol{u}_1^\top, \boldsymbol{u}_2^\top, \dots, \boldsymbol{u}_N^\top\right]^\top \in \mathbb{R}^{2N}$. The layer-wise degrees of freedom are retained independently so that temperature discontinuities may be captured at imperfect contacts.

Using the standard Galerkin procedure,

$$\boldsymbol{A}(\overline{\boldsymbol{u}})\dot{\boldsymbol{u}} = \left[\boldsymbol{B}^b(\overline{\boldsymbol{u}}) + \boldsymbol{B}^c\right]\boldsymbol{u} + \boldsymbol{f}(t), \tag{4}$$

where $\overline{\boldsymbol{u}} = \left[\bar{u}_1, \dots, \bar{u}_N\right]^\top$ contains the layer-average temperatures, $\boldsymbol{A} \in \mathbb{R}^{2N\times 2N}$ is the heat-capacity matrix, $\boldsymbol{B}^b \in \mathbb{R}^{2N\times 2N}$ is the bulk-conduction matrix, $\boldsymbol{B}^c \in \mathbb{R}^{2N\times 2N}$ accounts for imperfect thermal contacts, and $\boldsymbol{f} \in \mathbb{R}^{2N}$ contains the contributions from the prescribed boundary conditions.

$$\boldsymbol{A}(\overline{\boldsymbol{u}})\dot{\boldsymbol{u}} = \boldsymbol{B}(\overline{\boldsymbol{u}})\boldsymbol{u} + \boldsymbol{F}(t) \tag{5}$$

where $\boldsymbol{A}, \boldsymbol{B} \in \mathbb{R}^{2N\times 2N}$ are block diagonals containing the mass and stiffness matrices for each element, and $\boldsymbol{F}$ is the forcing vector upholding the boundary condition at each edge. In a TPS system with perfect contacts, these components are solely products of the choice in basis function; for LBFs, they are defined as

$$\boldsymbol{A}_i(\bar{u}) = \rho c_p(\bar{u}) \frac{L}{6} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \tag{6a}$$

$$\boldsymbol{B}_i(\bar{u}) = \frac{\mathrm{k}(\bar{u})}{L} \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \tag{6b}$$

$$\boldsymbol{F} = \int_{\partial\Omega} \boldsymbol{\phi} q_b \, d\Gamma \tag{6c}$$

where $\boldsymbol{A}_i, \boldsymbol{B}_i$ are the i-th blocks of $\boldsymbol{A}, \boldsymbol{B}$ respectively, and $\boldsymbol{F}$ is defined for the entire domain via its vector of LBFs, $\boldsymbol{\phi}$.

For a TPS with imperfect contacts, the bulk stiffness matrix in **??** no longer captures the total governing conductance of the TPS system, and thus must also account for the thermal resistances introduced by the imperfect contacts through the following heat interface heat flux condition,

$$\boldsymbol{B} = \underbrace{\boldsymbol{B}^b}_{\text{Satisfied}} + \underbrace{\boldsymbol{B}^c}_{\text{Missing}} \tag{7}$$

where the superscripts b and c differentiate the stiffness matrices that capture the *bulk* and *contact* heat fluxes, respectively. The choice in modeling the contact stiffness arises from the heat flux law

$$q = \frac{T_l - T_r}{R_x} \tag{8}$$

where the flux across a given element is defined by the change in temperature between the left and right sides over its thermal resistance. Leveraging the relationship of conductance and resistance, or

$$C = \frac{1}{R_x} \tag{9}$$

Eq. (8) can be rewritten as

$$q = C\left(T^- - T^+\right) \tag{10}$$

where T^- and T^+ represent the left and right sides of a given material block, respectively. Finally, generalizing for the full set of interactions between two locations on either side of a contact, the contact stiffness becomes

$$\boldsymbol{B}^c_j = C_j \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \tag{11}$$

for the j-th contact, where $j = i - 1$ due to the linearity in alignment of the elements (due to the one-dimensional assumption). The contact matrix can also be organized into a block diagonal matrix of shape $\boldsymbol{B}^c \in \mathbb{R}^{2M \times 2M}$, whose general structure is

$$\boldsymbol{B}^c = \begin{bmatrix} 0 & 0 & \cdots & 0 & 0 \\ 0 & \boldsymbol{B}^c_1 & \cdots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \cdots & \boldsymbol{B}^c_j & 0 \\ 0 & 0 & \cdots & 0 & 0 \end{bmatrix} \tag{12}$$

as there is no contact interface on the edges of the TPS.

Expanding Eq. (5) to now encompass all aspects of the distant contact problem yields

$$\boldsymbol{A}(\overline{\boldsymbol{u}})\dot{\boldsymbol{u}} = \boldsymbol{B}^b(\overline{\boldsymbol{u}})\boldsymbol{u} + \boldsymbol{B}^c\boldsymbol{u} + \boldsymbol{F}(t) \tag{13}$$

where $\boldsymbol{B}^c$ is not a function of average temperature or time, as conductance of an imperfect contact is assumed non-zero and constant.

2.4 Reduced-Physics Model

Next, the *lumped capacitance model* (LCM) under imperfect TPS layer contacts is presented as the RPM for the TPS. In this formulation, each layer is modeled as a lumped mass with a time-varying average temperature, where TCRs introduce additional physics such as intra-layer temperature discontinuities that must be modeled. To handle both bulk and contact heat transfer, the following formulation introduces the decomposition of the TPS layers into *bulk* and *contact* sub-domains, serving as the basis for subsequent coarse-graining derivations, TCR identifiability, and PIROM formulations.

Bulk Temperature Dynamics Leveraging our previous work in Ref. [VargasVenegas2026], the *bulk* temperature dynamics of the LCM is derived at the component level from a point of view of energy balance,

$$\int_{\Omega_i^b} \rho c_{p,i}(\bar{u})\dot{\bar{u}}_i d\Omega = \sum_{j\in\mathcal{N}_i^b} \int_{e_{ij}} q_{ij}^b d\Gamma + \int_{e_{iq}} q_b d\Gamma + \int_{e_{iT}} q_T d\Gamma \tag{14a}$$

$$\bar{A}_i\dot{\bar{u}}_i = \sum_{j\in\mathcal{N}_i^b} \int_{e_{ij}} q_{ij}^b d\Gamma + \int_{e_{iq}} q_b d\Gamma + \int_{e_{iT}} q_T d\Gamma,\ i = 1, 2, \ldots, N_B \tag{14b}$$

where $\bar{A}_i(\bar{u}_i) = \int_{\Omega_i^b} \rho c_{p,i}(\bar{u})d\Omega$, and q_{ij}^b, q_{ij}^c, q_{ib}, and q_{iT} correspond to the heat flux contributions due to neighboring elements with perfect (e_{ij}) contact, imperfect (Γ_{ij}) contact, Neumann boundary conditions (e_{iq}), and Dirichlet boundary conditions (e_{iT}), respectively. Moreover, $\mathcal{N}_i^b$ and $\mathcal{N}_i^c$ are the sets of neighboring elements across perfect and imperfect interfaces, respectively.

Interface Temperature Dynmaics Two equivalent approaches are presented in this work to formulate the temperature jump dynamics: (i) coarse-graining the DG-FEM model and applying a *zero-energy storage assumption* for the contact sub-domains as shown in [APPENDIX], and (ii) leveraging energy conservation at the interface level. For brevity, the second approach is adopted here, while the first approach is demonstrated in the [APPENDIX]. By energy conservation, *the bulk and contact heat fluxes are equal* at the imperfect interfaces, leading to the following algebraic constraint,

$$q_r^b(t) = \frac{\bar{u}^- - \bar{u}^+}{S_r(\bar{\mathbf{u}}) + R_r} = \frac{(\bar{u}^- - \bar{u}^+) - \Delta\bar{w}_r}{S_r(\bar{\mathbf{u}})} = g_r\Delta\bar{w}_r = q_r^c(t) \tag{15}$$

where $\Delta\bar{w}_r$ for $r = 1, 2, \ldots, N_\Gamma$ are contact temperature jumps, and depend on the TCR values. The heat flux continuity algebraic constraints in Eq. (15) are differentiated with respect to time, i.e., DAE index reduction, such that for each interface,

$$(1 + g_r S_r(\bar{\mathbf{u}}))\,\Delta\dot{\bar{w}}_r = \left(\dot{\bar{u}}^- - \dot{\bar{u}}^+\right) - \frac{(\bar{u}^- - \bar{u}^+) - \Delta\bar{w}_r}{S_r(\bar{\mathbf{u}})}\dot{S}_r(\bar{\mathbf{u}}), \quad r = 1, 2, \ldots, N_\Gamma \tag{16}$$

Collecting equations from all imperfect interfaces, the following system of ODEs governs the average contact temperature jump,

$$[\mathbf{I} + \mathbf{G}\odot\mathbf{S}(\bar{\mathbf{u}})]\,\Delta\dot{\bar{\mathbf{w}}} = \mathbf{D}\dot{\bar{\mathbf{u}}} - \mathbf{G}\left(\Delta\bar{\mathbf{w}}\odot\dot{\mathbf{S}}(\bar{\mathbf{u}})\right) \tag{17}$$

where the state $\Delta\bar{\mathbf{w}} \in \mathbb{R}^{N_\Gamma}$, conductance $\mathbf{G} \in \mathbb{R}^{N_\Gamma\times N_\Gamma}$ and bulk resistance $\mathbf{S}(\bar{\mathbf{u}}) \in \mathbb{R}^{N_\Gamma\times N_\Gamma}$ are defined as,

$$\mathbf{G} = \mathrm{diag}(g_1, g_2, \ldots, g_{N_\Gamma}) \tag{18a}$$

$$\mathbf{S}(\bar{\mathbf{u}}) = \mathrm{diag}(S_1(\bar{\mathbf{u}}), S_2(\bar{\mathbf{u}}), \ldots, S_{N_\Gamma}(\bar{\mathbf{u}})) \tag{18b}$$

while $\mathbf{D} \in \mathbb{R}^{N_\Gamma \times N_C}$ is the difference matrix that computes the lumped temperature differences across the interfaces, and $\odot$ denotes the Hadamard product.

Reduced-Physics Model Formulation At component i, the following assumptions are employed: (i) the bulk $\bar{u}_i$ and contact $\bar{w}_i$ average temperatures are uniform and equal to the spatial average over the their respective Ω_i^b and Ω_i^c sub-domains in component Ω_i, (ii) for neighboring components i and j under perfect contact, the bulk heat flux is defined as $q_{ij}^b = g_{ij}(\bar{\mathbf{u}})(\bar{u}_i - \bar{u}_j)$. The term $g_{ij}(\bar{\mathbf{u}}) = 1/R_{ij}(\bar{\mathbf{u}})$ is the effective thermal conductance, with $R_{ij} = R_i + R_j$ and $R_{ij}(\bar{\mathbf{u}}) = \ell/k(\bar{\mathbf{u}})A$, where ℓ is the distance between lumped masses, $k(\bar{\mathbf{u}})$ is the temperature-dependent thermal conductivity, and A is the cross-sectional area of the interface. Similarly, the heat flux due to the Dirichlet boundary condition is given by $q_{iT} = g_{iT}(\bar{u}_i - \bar{u}_T)$. Now for component i, and noting that $\mathcal{N}_i^b = \mathcal{N}_i^u \cup \mathcal{N}_i^w$, Eq. (14b) may be written as,

$$\bar{A}_i(\bar{u}_i)\dot{\bar{u}}_i = \sum_{j\in\mathcal{N}_i^u} \int_{e_{ij}} g_{ij}(\bar{\mathbf{u}})(\bar{u}_i - \bar{u}_j)\,de + \sum_{j\in\mathcal{N}_i^w} \int_{e_{ij}} g_{ij}(\bar{\mathbf{u}})(\bar{u}_i - \bar{w}_j)\,de + \int_{e_{iq}} q_b de + \int_{e_{iT}} q_T de \tag{19a}$$

$$= \sum_{j\in\mathcal{N}_i^b\cup\{T\}} g_{ij}(\bar{\mathbf{u}})\,|e_{ij}|\,\bar{u}_i - \sum_{j\in\mathcal{N}_i^u} g_{ij}(\bar{\mathbf{u}})\,|e_{ij}|\,\bar{u}_j - \sum_{j\in\mathcal{N}_i^w} g_{ij}(\bar{\mathbf{u}})\,|e_{ij}|\,\bar{w}_j + |e_{iq}|\,\bar{q}_b - g_{iT}(\bar{\mathbf{u}})\,|e_{iT}|\,\bar{T}_b \tag{19b}$$

$$= \sum_{j\in\mathcal{N}_i^b\cup\{T\}} \bar{B}_{ij}^i \bar{u}_i + \sum_{j\in\mathcal{N}_i^u} \bar{B}_{ij}^j \bar{u}_j + \sum_{j\in\mathcal{N}_i^w} \bar{B}_{ij}^j \bar{w}_j + \bar{f}_i^{(b)}(t) \tag{19c}$$

or, by leveraging heat flux continuity from Eq. (15),

$$\left(\int_{\Omega_i^b} \rho c_{p,i}(\bar{u}) d\Omega\right) \dot{\bar{u}}_i = \left(\sum_{j\in\mathcal{N}_i^b} \int_{e_{ij}} g_{ij}(\bar{u}_i - \bar{u}_j)\,d\Gamma\right) \tag{20}$$

$$+ \sum_{j\in\mathcal{N}_i^c} \left(\int_{\Gamma_{ij}} g_{x,ij} \Delta\bar{w}_j d\Gamma\right) + \int_{e_{iq}} q_b d\Gamma + \int_{e_{iT}} q_T d\Gamma \tag{21}$$

Collecting N_B bulk and N_Γ interface terms, the system of equations governing the coarse-grained temperature dynamics of the TPS with imperfect thermal contacts is written as the following coupled system of ODEs,

$$\overline{\boldsymbol{\mathcal{A}}}(\mathbf{s})\,\dot{\mathbf{s}} = \overline{\boldsymbol{\mathcal{B}}}(\mathbf{s}) + \overline{\boldsymbol{\mathcal{F}}}(t) \tag{22}$$

where,

$$\overline{\boldsymbol{\mathcal{A}}} = \begin{bmatrix} \bar{\mathbf{A}}(\bar{\mathbf{u}}) & \mathbf{0} \\ -\mathbf{D} & \mathbf{I} + \mathbf{G}\odot\mathbf{S}(\bar{\mathbf{u}}) \end{bmatrix}, \quad \overline{\boldsymbol{\mathcal{B}}} = \begin{bmatrix} \bar{\mathbf{B}}(\bar{\mathbf{u}}) & \bar{\mathbf{K}}(\bar{\mathbf{u}}) \\ \mathbf{0} & -\mathbf{G}\odot\dot{\mathbf{S}}(\bar{\mathbf{u}}) \end{bmatrix}, \quad \overline{\boldsymbol{\mathcal{F}}}(t) = \begin{bmatrix} \bar{\mathbf{f}}(t) \\ \mathbf{0} \end{bmatrix} \tag{23}$$

and where $\mathbf{s} = [\bar{\mathbf{u}}, \Delta\bar{\mathbf{w}}]^\top \in \mathbb{R}^{N_C+N_\Gamma}$, and $\overline{\boldsymbol{\mathcal{A}}}, \overline{\boldsymbol{\mathcal{B}}} \in \mathbb{R}^{(N_C+N_\Gamma)\times(N_C+N_\Gamma)}$ define the RPM. Note that the $\overline{\boldsymbol{\mathcal{A}}}$ and $\overline{\boldsymbol{\mathcal{B}}}$ matrices include temperature dependence that represents the primary source of nonlinearities for the problems considered in this work.

3 Physics-Infused Reduced-Order Modeling

Following our previous work in Ref.[VargasVenegas2026], the PIROM formulation starts by mathematically connecting the FOM to the RPM.

The formulation of PIROM for TPS starts by mathematically connecting the FOM to the RPM. This connection is formally established through a coarse-graining procedure, which systematically reduces the spatial resolution of the DG-FEM to discover the lifted LCM model with contact temperature jump dynamics. Additionally, the coarse-graining procedure pinpoints residual terms in the RPM that facilitate the design of a data-driven memory closure to account for discarded spatial resolution.

3.1 Deriving the Reduced-Physics Model via Coarse-Graining

Following our previous work in Ref. [VargasVenegas2026], the LCM can be viewed as a zeroth-order (i.e., constant basis functions) DG-FEM with an extremely coarse mesh partition, and where the element-wise penalty factors become the conductance. The presence of imperfect interfaces motivates the modification of the coarse-graining procedure to explicitly introduce an additional dynamical system that is parametrized by TCRs, which is not present in the standard LCM formulation. The following discussion presents the derivation of the modified LCM dynamics via coarse-graining, and the PIROM formulation for residual dynamics.

3.1.1 Coarse-Graining of States

Consider a DG-FEM formulation as in Eq. (3) with M elements, N_C components and N_Γ interfaces; clearly, $M \gg N_C, N_\Gamma$. Let component $\Omega_i = \Omega_i^b \cup \Omega_i^c$ share a boundary with the imperfect interface Γ_r, i.e., $\partial\Omega_i^c \cap \Gamma_r \neq \emptyset$ with $\Omega_i^c \subset \Omega_i$, and define the sets of indices $\mathcal{V}_i = \{j \mid E_j \subset \Omega_i^b\}, \quad i = 1, 2, \ldots, N_B$ and $\mathcal{F}_i = \{j \mid \partial W_j \cap \Gamma_r \neq \emptyset,\ W_j \subset \Omega_i^c\}, \quad i = 1, 2, \ldots, N, r = 1, 2, \ldots, N_C$ for elements in the bulk Ω_i^b and contact Ω_i^c sub-domains, respectively. Introduce the following states $\mathbf{w} = [\mathbf{w}^-, \mathbf{w}^+]^\top \in \mathbb{KP}$, where $\mathbf{w}^-$ and $\mathbf{w}^+$ are the states of elements on the negative and positive sub-domains conforming and interface, respectively. Note that $M = B + K$, and $\mathbf{w}^- \in \mathbb{R}^{K^-P}$ and $\mathbf{w}^+ \in \mathbb{R}^{K^+P}$, with $Ks = K^- + K^+$ and where K^- and K^+ denote the total number of elements on the negative and positive sides of imperfect interfaces, respectively.

Under such variable construction, the average temperatures on some bulk component Ω_i^b and interface Γ_r are given by,

$$\bar{u}_i = \frac{1}{|\Omega_i^b|} \sum_{m\in\mathcal{V}_i} \int_{E_m} \boldsymbol{\phi}_m^\top \mathbf{u}_m dE = \frac{1}{|\Omega_i^b|} \sum_{m\in\mathcal{V}_i} |E_m|\, \boldsymbol{\varphi}_m^{j\top} \mathbf{u}_m, \quad i = 1, \ldots, N_C \tag{24a}$$

$$\bar{w}_r^-(t) = \frac{1}{\left|\Omega_r^{c,-}\right|} \sum_{m\in\mathcal{F}_r^-} \int_{E_m} \boldsymbol{\phi}_m^\top \mathbf{w}_m^- dE = \frac{1}{\left|\Omega_r^{c,-}\right|} \sum_{m\in\mathcal{F}_r^-} |E_m|\, \boldsymbol{\psi}_m^{r-} \mathbf{w}_m^-, \quad r = 1, 2, \ldots, N_\Gamma \tag{24b}$$

$$\bar{w}_r^+(t) = \frac{1}{\left|\Omega_r^{c,+}\right|} \sum_{m\in\mathcal{F}_r^+} \int_{E_m} \boldsymbol{\phi}_m^\top \mathbf{w}_m^+ dE = \frac{1}{\left|\Omega_r^{c,+}\right|} \sum_{m\in\mathcal{F}_r^+} |E_m|\, \boldsymbol{\psi}_m^{r+} \mathbf{w}_m^+, \quad r = 1, 2, \ldots, N_\Gamma \tag{24c}$$

where the orthogonal basis functions $\boldsymbol{\varphi}_i^j, \boldsymbol{\psi}_m^{j\pm} = [1, 0, \ldots, 0] \in \mathbb{R}^P$ if $i \in \mathcal{V}_j$ and $m \in \mathcal{F}_j^\pm$, respectively, and are zero otherwise. With the average temperatures $\{\bar{u}_i\}_{i=1}^{N_C}$, $\{\bar{w}_i^-\}_{i=1}^{N_\Gamma}$, and $\{\bar{w}_i^+\}_{i=1}^{N_\Gamma}$, the DG-FEM states of elements on domains Ω_i^b, $\Omega_r^{c,-}$, and $\Omega_r^{c,+}$ are written as,

$$\mathbf{u}_i = \sum_{k=1}^{N_C} \boldsymbol{\varphi}_i^k \bar{u}_k + \delta\mathbf{u}_i, \quad i \in \mathcal{V}_j, \quad j = 1, 2, \ldots, B \tag{25a}$$

$$\mathbf{w}_{ir}^- = \sum_{k=1}^{N_\Gamma} \boldsymbol{\psi}_k^{r-} \bar{w}_k^- + \delta \mathbf{w}_{ir}^-, \quad i \in \mathcal{F}_r^-, \quad r = 1, 2, \dots, N_\Gamma \tag{25b}$$

$$\mathbf{w}_{ir}^+ = \sum_{k=1}^{N_\Gamma} \boldsymbol{\psi}_k^{r+} \bar{w}_k^+ + \delta \mathbf{w}_{ir}^+, \quad i \in \mathcal{F}_r^+, \quad r = 1, 2, \dots, N_\Gamma \tag{25c}$$

where $\boldsymbol{\varphi}_i^k, \boldsymbol{\psi}_i^{k\pm} = [1, 0, \dots, 0]^\top \in \mathbb{R}^P$ if $i \in \mathcal{V}_k$ and $\mathcal{F}_r^\pm$, respectively.

The Eqs. (25a) to (25c) are collected in matrix form,

$$\mathbf{u} = \boldsymbol{\Phi}\bar{\mathbf{u}} + \delta\mathbf{u}, \quad \bar{\mathbf{u}} = \boldsymbol{\Phi}^\dagger \mathbf{u} \tag{26a}$$

$$\mathbf{w} = \boldsymbol{\Pi}\bar{\mathbf{w}} + \delta\mathbf{w}, \quad \bar{\mathbf{w}} = \boldsymbol{\Pi}^\dagger \mathbf{w} \tag{26b}$$

where $\boldsymbol{\Phi} \in \mathbb{R}^{BP \times N_B}$ is a matrix of $B \times N_B$ blocks, with the (i, j)-th block as $\boldsymbol{\varphi}_i^j$, $\boldsymbol{\Phi}^\dagger \in \mathbb{R}^{N_C \times BP}$ is the left inverse of $\boldsymbol{\Phi}$, with the (i, j)-th block as $\boldsymbol{\varphi}_i^{j\dagger} = \frac{|E_i|}{|\Omega_j|} \boldsymbol{\varphi}_i^{jT}$ and $\delta\mathbf{u}$ is the collection of deviations from the average temperature. Moreover,

$$\boldsymbol{\Pi} = \begin{bmatrix} \mathbf{M} & \mathbf{0} \\ \mathbf{0} & \mathbf{N} \end{bmatrix} \in \mathbb{R}^{KP \times N_C}, \quad \boldsymbol{\Pi}^\dagger = \begin{bmatrix} \mathbf{M}^\dagger & \mathbf{0} \\ \mathbf{0} & \mathbf{N}^\dagger \end{bmatrix} \in \mathbb{R}^{2N_\Gamma \times KP} \tag{27}$$

where $\mathbf{M} \in \mathbb{R}^{K^- P \times N_\Gamma}$ and $\mathbf{N} \in \mathbb{R}^{K^+ P \times N_\Gamma}$ correspond to matrices that project the average interface temperatures $\bar{\mathbf{w}} = [\bar{\mathbf{w}}^-, \bar{\mathbf{w}}^+] \in \mathbb{R}^{2N_\Gamma}$ to the full-order interface DG-FEM state. Therefore, $\boldsymbol{\Pi} \in \mathbb{R}^{KP \times 2N_\Gamma}$ is a matrix of $K \times 2N_\Gamma$ blocks with the (i, j)-th block as $\boldsymbol{\psi}_i^j$, $\boldsymbol{\Pi}^\dagger \in \mathbb{R}^{2N_\Gamma \times KP}$ is the left inverse of $\boldsymbol{\Pi}$ with the (i, j)-th block as $\boldsymbol{\psi}_i^{j\dagger} = \frac{|E_i|}{|\Omega_j^c|} \boldsymbol{\psi}_i^{j\top}$ and $\delta\mathbf{w}$ is the collection of deviations from the interfacial average temperatures. By definition, the matrices defined above satisfy,

$$\begin{aligned} \boldsymbol{\Phi}^\dagger \boldsymbol{\Phi} &= \mathbf{I}, & \boldsymbol{\Phi}^\dagger \delta\mathbf{u} &= \mathbf{0} \\ \boldsymbol{\Pi}^\dagger \boldsymbol{\Pi} &= \mathbf{I}, & \boldsymbol{\Pi}^\dagger \delta\mathbf{w} &= \mathbf{0} \end{aligned}$$

For brevity, the states in Eq. (26b) are collectively written as,

$$\mathbf{x} = \mathbf{X}\bar{\mathbf{x}} + \delta\mathbf{x}, \quad \bar{\mathbf{x}} = \mathbf{X}^\dagger \mathbf{x} \tag{29}$$

where $\mathbf{x} = [\mathbf{u}, \mathbf{w}]^\top \in \mathbb{R}^{MP}$, $\mathbf{X} \in \mathbb{R}^{MP \times (N_B + N_C)}$, and $\mathbf{X}^\dagger \in \mathbb{R}^{(N_B + N_C) \times MP}$.

3.1.2 Coarse-Graining of Dynamics

Next, the coarse-graining procedure follows our previous work in Ref. [VargasVenegas2026], where the higher-order temperature modes are neglected in the DG-FEM dynamics, leading to a spatially averaged transient modeling of the TPS temperature dynamics. The presence of imperfect thermal contacts introduces additional modeling considerations to account for temperature discontinuities across TPS layers. The following discussion summarizes the critical results from coarse-graining, while detailed derivations are provided in Appendix [X].

Consider a function of the states in the form $\mathbf{M}(\mathbf{x})\, \mathbf{g}(\mathbf{x})$ with $\mathbf{x} = [\mathbf{u}, \mathbf{w}]^\top \in \mathbb{R}^{MP}$ and $M = B + K$, where $\mathbf{g}(\mathbf{x}) : \mathbb{R}^{MP} \to \mathbb{R}^{MP}$ is a vector-valued function, and $\mathbf{M}(\mathbf{x}) : \mathbb{R}^{MP} \to \mathbb{R}^{p \times MP}$ is a matrix-valued function with an arbitrary dimension p. Define the projection matrices $\mathbf{P} = \mathbf{X}\mathbf{X}^\dagger$ where,

$$\mathbf{X} = \operatorname{diag}(\boldsymbol{\Phi}, \boldsymbol{\Pi}) \in \mathbb{R}^{MP \times (N_B + N_C)} \tag{30a}$$

$$\mathbf{X}^\dagger = \mathrm{diag}\left(\mathbf{\Phi}^\dagger, \mathbf{\Pi}^\dagger\right) \in \mathbb{R}^{(N_B+N_C)\times MP} \tag{30b}$$

and the projection operator,

$$\begin{aligned}
\mathcal{P}\left[\mathbf{M}(\mathbf{x})\mathbf{g}(\mathbf{x})\right] &= \mathbf{M}(\mathbf{P}\mathbf{x})\mathbf{P}\mathbf{g}(\mathbf{P}\mathbf{x}) && (31a)\\
&= \mathbf{M}\left(\mathbf{X}\mathbf{X}^\dagger\mathbf{x}\right)\mathbf{X}\mathbf{X}^\dagger\mathbf{g}\left(\mathbf{X}\mathbf{X}^\dagger\right) && (31b)\\
&= \mathbf{M}\left(\mathbf{X}\bar{\mathbf{x}}\right)\mathbf{X}\mathbf{X}^\dagger\mathbf{g}\left(\mathbf{X}\bar{\mathbf{x}}\right) && (31c)
\end{aligned}$$

so that the resulting function depends only on the resolved variables $\bar{\mathbf{x}} = \left[\bar{\mathbf{u}}, \bar{\mathbf{w}}^-, \bar{\mathbf{w}}^+\right]^{N_B+N_C}$; note that an averaged temperature difference $\Delta\bar{\mathbf{w}} = \bar{\mathbf{w}}^- - \bar{\mathbf{w}}^+$ may be computed between two contact sub-domains conforming an imperfect interface, which becomes useful for subsequent TCR identifications. Correspondingly, the residual operator is $\mathcal{Q} = \mathcal{I} - \mathcal{P}$ so that $\mathcal{Q}\left[\mathbf{M}(\mathbf{x})\mathbf{g}(\mathbf{x})\right] = \mathbf{M}(\mathbf{x})\mathbf{g}(\mathbf{x}) - \mathcal{P}\left[\mathbf{M}(\mathbf{x})\mathbf{g}(\mathbf{x})\right]$. When the function is not separable, the projection operator is simply given by $\mathcal{P}\left[\mathbf{g}(\mathbf{x})\right] = \mathbf{g}\left(\mathbf{X}\bar{\mathbf{x}}\right)$.

Subsequently, the operators defined above are applied to coarse-grain the dynamics. First, the DG-FEM model in Eq. (3) is written as,

$$\dot{\mathbf{x}} = \boldsymbol{\mathcal{A}}^{-1}\left[\boldsymbol{\mathcal{B}}(\mathbf{x})\mathbf{x} + \boldsymbol{\mathcal{F}}(t)\right] = \mathbf{r}(\mathbf{x}, t) \tag{32}$$

Multiplying both sides by $\mathbf{X}^\dagger$ produces,

$$\mathbf{X}^\dagger\dot{\mathbf{x}} = \mathbf{X}^\dagger\left(\mathbf{X}\dot{\bar{\mathbf{x}}} + \delta\dot{\mathbf{x}}\right) = \dot{\bar{\mathbf{x}}} = \mathbf{X}^\dagger\mathbf{r}(\mathbf{x}, t) \tag{33}$$

and applying the projection and residual operator yields,

$$\begin{aligned}
\dot{\bar{\mathbf{x}}} &= \mathcal{P}\left[\mathbf{X}^\dagger\mathbf{r}(\mathbf{x}, t)\right] + \mathcal{Q}\left[\mathbf{X}^\dagger\mathbf{r}(\mathbf{x}, t)\right] && (34a)\\
&= \mathbf{r}^{(1)}(\bar{\mathbf{x}}, t) + \mathbf{r}^{(2)}(\mathbf{x}, t) && (34b)
\end{aligned}$$

where $\mathbf{r}^{(1)}(\bar{\mathbf{x}}, t)$ is the resolved dynamics, which depends only on $\bar{\mathbf{x}}$, and $\mathbf{r}^{(2)}(\mathbf{x}, t)$ is the residual dynamics, which depends on the full-order state $\mathbf{x}$. Detailed derivations of the resolved $\mathbf{r}^{(1)}$ and residual $\mathbf{r}^{(2)}$ dynamics are provided in Appendix [X], and the key results are stated below.

First, the resolved dynamics becomes exactly the LCM, with the temperature jump dynamics formulated as a *zero-energy storage* assumption on the contact sub-domain component of the DG-FEM dynamics. Specifically, the coarse-grained mass matrices are given by,

$$\begin{aligned}
\bar{\mathbf{A}} &= \mathbf{W}_u\left(\mathbf{\Phi}^\dagger\mathbf{A}\left(\mathbf{\Phi}\bar{\mathbf{u}}\right)^{-1}\mathbf{\Phi}\right)^{-1} && (35a)\\
\bar{\mathbf{D}} &= \mathbf{W}_w\left(\mathbf{\Pi}^\dagger\mathbf{D}\left(\mathbf{\Pi}\bar{\mathbf{w}}\right)^{-1}\mathbf{\Pi}\right)^{-1} && (35b)
\end{aligned}$$

where $\mathbf{W}_u \in \mathbb{R}^{N_B\times N_B}$ and $\mathbf{W}_w \in \mathbb{R}^{N_C\times N_C}$ are diagonal weight matrices such that $\left[\mathbf{W}_u\right]_i = |\mathcal{V}_k|$ if $i \in \mathcal{V}_k$, and $\left[\mathbf{W}_c\right]_i = |\mathcal{F}_k|$ if $i \in \mathcal{F}_k$. Similarly, the coarse-grained stiffness matrices are,

$$\begin{aligned}
\bar{\mathbf{B}} &= \mathbf{W}_u\mathbf{\Phi}^\dagger\mathbf{B}\left(\mathbf{\Phi}\bar{\mathbf{u}}\right)\mathbf{\Phi} && (36a)\\
\bar{\mathbf{C}} &= \mathbf{W}_u\mathbf{\Phi}^\dagger\mathbf{C}\left(\mathbf{\Phi}\bar{\mathbf{u}}\right)\mathbf{\Pi} && (36b)\\
\bar{\mathbf{E}} &= \mathbf{W}_w\mathbf{\Pi}^\dagger\mathbf{E}\left(\mathbf{\Phi}\bar{\mathbf{u}}\right)\mathbf{\Phi} && (36c)\\
\bar{\mathbf{F}} &= \mathbf{W}_w\mathbf{\Pi}^\dagger\mathbf{F}\left(\mathbf{\Phi}\bar{\mathbf{u}}\right)\mathbf{\Pi} && (36d)
\end{aligned}$$

where

3.2 Formulation of Reduced-Order Model

The Mori-Zwanzig (MZ) formalism is an operator-projection technique used to derive ROMs from high-dimensional dynamical systems, especially in statistical mechanics and fluid dynamics [Parish2017,Stinis2017,Parish2025]. The MZ formalism enables an exact reformulation of the FOM in terms of a subset of resoled states, with unresolved dynamical terms appearing as residual

The proposed PIROM is subsequently developed based on such reformulation.

4 Bulk-only Mori–Zwanzig Memory Closure

The purpose of the closure is to account for the unresolved dynamics in the LCM, while leaving the contact-jump dynamics unchanged. Accordingly, the

Accordingly, write the Projected Generalized Langevin Equation (PGLE) for the lumped-temperature dynamics,

$$A(\mathbf{u})\dot{\mathbf{u}} = \sum_{i=1}^{N-1} \frac{1}{S_i(\mathbf{u})} \left(\mathbf{K}_i u + \mathbf{L}_i \Delta\boldsymbol{\theta}\right) + \mathbf{f}(t) + \int_{t_0}^{t} \boldsymbol{\kappa}(t, s, u, \mathbf{f}; \Theta) ds \tag{37}$$

Now, ensure the designed memory modifies only the bulk conductances, i.e.,

$$\boldsymbol{\kappa} \in \operatorname{span}\left\{\mathbf{K}_1 u(t), \mathbf{K}_2 u(t), \mathbf{K}_3 u(t), \mathbf{L}_1 \Delta\boldsymbol{\theta}_1(t), \mathbf{L}_2 \Delta\boldsymbol{\theta}_2(t), \mathbf{L}_3 \Delta\boldsymbol{\theta}_3(t)\right\} \tag{38}$$

which leads to,

$$\boldsymbol{\kappa}(t, s, u, \mathbf{f}; \Theta) \approx \sum_{j=1}^{m} \kappa_j(t, s, u, \mathbf{f}; \Theta) \boldsymbol{\mathcal{P}}_j \tag{39}$$

where,

$$\boldsymbol{\mathcal{P}}_j = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \mathbf{p}_{ij} \tag{40}$$

Now, define the memory variables as,

$$\beta_j(t) = \int_{t_0}^{t} \kappa_j(t, s, u, \mathbf{f}; \Theta) ds \tag{41}$$

then,

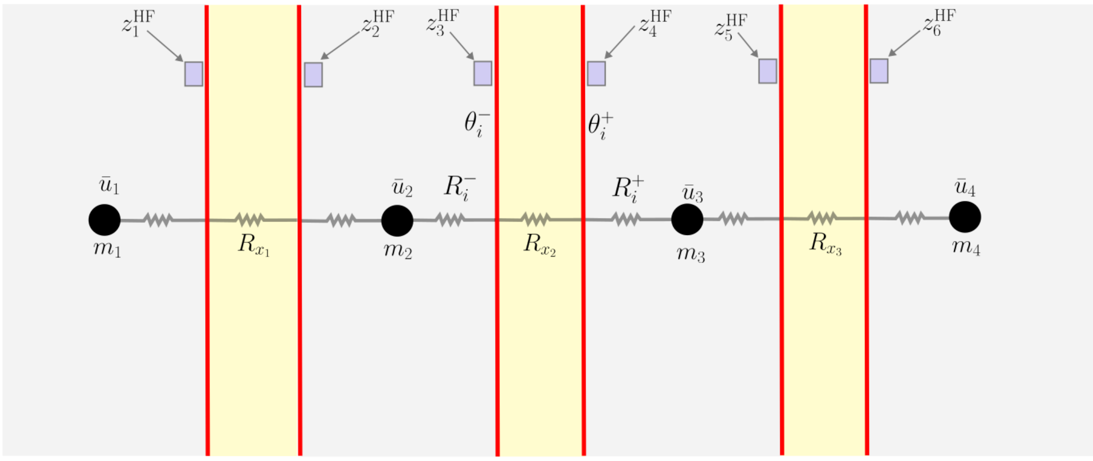
$$\int_{t_0}^{t} \boldsymbol{\kappa}(t, s, u, \mathbf{f}; \Theta) ds = \sum_{j=1}^{m} \beta_j(t) \boldsymbol{\mathcal{P}}_j \tag{42}$$

It follows that,

$$\sum_{j=1}^{m} \beta_j(t) \boldsymbol{\mathcal{P}}_j = \sum_{j=1}^{m} \beta_j(t) \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \mathbf{p}_{ij} = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \left(\sum_{j=1}^{m} \mathbf{p}_{ij} \beta_j(t) \right) = \sum_{i=1}^{N-1} \frac{1}{S_i(u)} \mathbf{P}_i \boldsymbol{\beta} \tag{43}$$

Then, the memory correction to the LCM dynamics leads to the following modified lumped-temperature dynamics,

$$A(\mathbf{u})\dot{\mathbf{u}} = \sum_{i=1}^{N-1} \frac{1}{S_i(\mathbf{u})} \left(\mathbf{K}_i u + \mathbf{L}_i \Delta\boldsymbol{\theta} + \mathbf{P}_i \boldsymbol{\beta}\right) + \mathbf{f}(t) \tag{44}$$



$$q_i^{(c)} = \frac{\theta_i^+ - \theta_i^-}{R_{x_i}}$$

Now, let,

$$\kappa_i\left(t, s, \mathbf{u}, \mathbf{f}; \Theta\right) = e^{-\lambda_j(t-s)}\left(q_j^\top \phi(\mathbf{u}(s)) + r_j^\top \psi(\mathbf{f}(s))\right) \tag{45}$$

such that the memory variables are given by,

$$\boldsymbol{\beta}(t) = \int_{t_0}^{t} e^{-\Lambda(t-s)}\left(Q^\top \phi(\mathbf{u}(s)) + R^\top \psi(\mathbf{f}(s))\right) ds \tag{46}$$

The hidden dynamics are now given by,

$$\dot{\boldsymbol{\beta}} = -\Lambda\boldsymbol{\beta} + Q^\top \phi(\mathbf{u}) + R^\top \psi(\mathbf{f}) \tag{47}$$

and in state-space form, the hidden dynamics are,

$$\dot{\boldsymbol{\beta}} = -\Lambda\boldsymbol{\beta} + Q^\top \phi(\mathbf{u}) + R^\top \psi(\mathbf{f}) \tag{48}$$

The full lifted and closed system is given by,

$$A(\mathbf{u})\dot{\mathbf{u}} = \sum_{i=1}^{N-1} \frac{1}{S_i(\mathbf{u})}\left(\mathbf{K}_i\mathbf{u} + \mathbf{L}_i\Delta\boldsymbol{\theta}\right) + \mathbf{f}(t) \tag{49a}$$

$$M(\mathbf{u}; R_x)\dot{\Delta\boldsymbol{\theta}} = \mathbf{D}\dot{\mathbf{u}} - G_x\left(\Delta\boldsymbol{\theta} \odot \dot{S}(\mathbf{u})\right) \tag{49b}$$

$$\dot{\boldsymbol{\beta}} = -\Lambda\boldsymbol{\beta} + Q^\top \phi(\mathbf{u}) + R^\top \psi(\mathbf{f}) \tag{49c}$$

$$\mathbf{z} = \mathbf{H}_u\mathbf{u} + \mathbf{H}_\theta\Delta\boldsymbol{\theta} + \mathbf{H}_{\boldsymbol{\beta}}\boldsymbol{\beta} \tag{49d}$$

$$\mathcal{J} = \alpha_1\mathcal{J}_{\mathrm{TCR}} + \alpha_2\mathcal{J}_\beta \tag{50}$$

5 Application to Thermal Protection Systems

In this section, the proposed PIROM approach is applied to modeling a four-layered hypersonic TPS with unknown contact resistances, and is compared against a representative physics-based model, i.e., FEMs, and data-driven surrogate model, i.e., a neural ordinary differential equation (NODE) model. The performance of the models is assessed in terms of their ability to infer the TCRs, and to predict the transient temperature dynamics under varying boundary conditions and material properties. The configuration of the TPS, problem parametrization, optimal model design for inference, and numerical results are presented next.

5.1 Problem Description and Parametrization

5.2 Model Selection

5.2.1 One-Dimensional Finite-Element Model

In Section **??**, the 1D-FEM was introduced as utilizing linear basis functions (LBFs). These LBFs have favorable behavior when put in weak form, yielding simple stiffness and mass matrices that captures the heat transfer within material blocks and the imperfect interfaces between said blocks. However, the 1D-FEM also has the added benefit of being a first-order model; consequently, by leveraging the pair of LBFs on each side of a respective material block, the temperature at any given sensor position within the block can be easily computed via a weighted average.

Consider the example presented in Fig. 2, where each of the sensors are perturbed from their respective interfaces by distance δ. The temperature of z_1^{HF} when the temperatures on the left and right side of the first material block are u_l and u_r respectively is then given by

$$u = u_l \cdot \frac{L-\delta}{L} + u_r \cdot \frac{\delta}{L} \tag{51}$$

where L is the thickness of the material block in question. By computing the weighted average of every sensor with respect to the two edge temperatures on either side, the sensor temperatures can be found regardless of the distance δ in which they are perturbed.

To examine the ability of the 1D-FEM in inferring the gap TCRs, the model will be tested in a range of experiments with two varying parameters: noise level and perturbation distance, δ. For varying noise level, the high fidelity sensor readings will be injected with random Gaussian noise to corrupt the temperature signals. These sensors will then be denoised using a weak form kernel ridge regression framework [**?**] prior to TCR training; the subsequent denoising should clean and smooth the signal, but may introduce some biases that could interfere with the inferred conductances. In response, several seeds will be evaluated for each experiment (i.e., noise level and sensor placement) — allowing for the analysis of the overall behavior of adding noise to the system, and neglecting the erratic residue of any single seed performance. Doing so should make the evolution of TCR inference error with respect to increasing noise level clear.

Additionally, six different sensor perturbation distances will be evaluated, or δ_i for $i = 1, 2, \ldots, 6$. These perturbations range from 0% - 20% of the block length L, with higher i values corresponding to more extreme displacements of the sensors away from their respective interfaces. By sweeping a range of sensor positions from perfect to extremely perturbed placement, the trend of how conductance inference error is affected by moving the sensors away from their interfaces will be clear.

The mean error across the ensemble is presented in Fig. 4, where the 1D-FEM model occupies the first row. It is immediately apparent that the dominant error mode of TCR inference is the sensor position, as error significantly increases as the sensor is perturbed further away from the interface. Such behavior is likely a byproduct of the LBFs, which can only accurately predict the sensor temperature readings within a small distance of the interfaces. Most of this error originates from the sensor in Acusil, whose low conductance produces an asymptotic curve that is difficult to approximate with LBFs alone. Similarly, the error increases slightly as the noise level increases, an artifact of the small biases introduced by the noise/denoise framework that shift the inferred TCR solution.

The standard deviation error for the same ensemble is illustrated in Fig. 5, where the noise level is a major source of variation. Observe the negligible STD of the no-noise case where all ensembles produce essentially identical solutions, while the moderate- and high-noise cases produce steadily increasing STDs as a result of the denoising. Since more extreme noises may introduce similarly behaving biases, different seeds can produce greatly varying conductance errors. Note that at further sensor positions the perturbation error dominates the that of the noise significantly, resulting in less STD as the sensor moves further and further away (but still higher mean error).

To summarize, the 1D-FEM performs exceptionally well when the sensor is near-perfectly placed with no significant noise. However, at large displacements or high noise levels this error rises quite significantly, an artifact of both the diminishing accuracy of the LBFs and the biases introduced by the denoising algorithm.

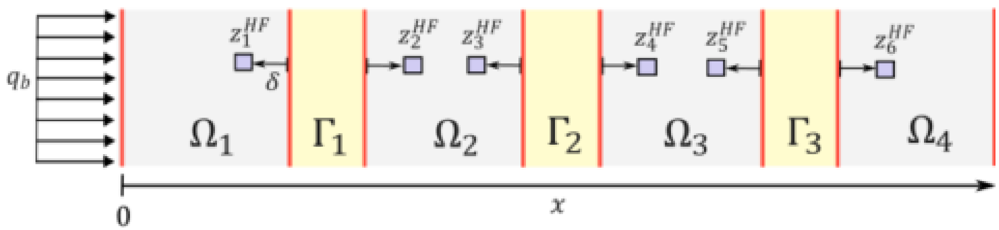


Figure 2: Experimental setup with arbitrary sensor displacement, δ, applied to each of the six sensors.

1. Discuss sensor placement and how it is done for this specific system (i.e. LBFs)

2. Tradeoff for C vs δ vs SNR (however, this might be best placed in the appendix with the main takeaways in the main document)

5.2.2 Lumped Capacitance Model

The lumped capacitance models (LCMs) from Section **??**, which are contrarily based in a finite-volume model, are rooted differently. Rather than leveraging LBFs to formulate a first-order model like the 1D-FEM, the LCMs operate as a zeroth-order model; that is, the only known values are the average block temperatures, and all other information must be subsequently deduced. Consequently, the approach of computing the temperature is more elaborate, consisting of recomputing effective conductances.

Consider the demonstrative experiment presented in Fig. 3, where a pair of sensors on either side of an interface are shown. By adjusting the δ sensor position, the effective resistance across a gap can be computed — by moving the sensor away from the interface, the calculation must not only account for the gap resistance, but also the resistance of the block between the sensor and the interface. Consequently, the corrected resistance ensures the flux is accurate, correcting the temperatures jump in temperature across the gap and ensuring the inferred conductance is accurate.

For perfect sensor placement, the resistance across the gap is simply R_{x_i}. However, as the sensor is moved away from the interface, one must account for the resistances in either material block for a proper IHCP. Thus, it is imperative to adjust the temperature jump across the gap; as heat flux is constant across the material blocks and gap interfaces, this is quite intuitive. Consider

$$q = \frac{\Delta T}{R_{\text{eff}}} \tag{52}$$

where the flux is given by the temperature jump across a gap with effective resistance R_{eff}. Since the flux across the interface itself will be equal to the flux through the material block and gap, the temperature difference at the new sensor positions can be computed through their relation. For $\Delta\theta$ and Δz representing the temperature difference for the perfect and perturbed sensor placement respectively, the relation is

$$\frac{\Delta z}{R_{x_1} + R_1^- + R_1^+} = \frac{\Delta\theta}{R_{x_i}} \tag{53}$$

which yields the corrected temperature jump as

$$\Delta z = \Delta\theta \left[1 + \frac{R_1^- + R_1^+}{R_{x_i}}\right] \tag{54}$$

This framework shows how finite-volume methodology can be leveraged to compute the temperatures at the new sensor locations. As an added sanity check, if the sensor placement is perfect, $R_i^- = R_i^+ = 0$, which would yield $\Delta z = \Delta\theta$.

Similarly to the 1D-FEM, the LCM will be examined on its ability to infer the gap TCRs. However, in addition to testing the noise levels and perturbation distance, the refinement of the model will also be analyzed. For instance, the most coarse LCM can have four elements, one for each block — this is denoted as LCM 4. However, the number of elements can be greater than four as well. Consider an LCM with six elements (i.e., LCM 6) which can be distributed to the four material blocks. A naive way of handling this is to place the two extra elements in random blocks, but a more structured framework is to place both extra blocks in Acusil. As Acusil has a significantly smaller conductivity — and thus a larger temperature gradient — the zeroth-order approximation would be the most inadequate here. Therefore, for LCM models with five or more elements, the extra elements will be isolated to the first material block where they can improve tracking of the Acusil temperature profile near the sensor. To supplement this framework, a spline can also be fit for the two elements nearest to the sensor in the Acusil layer, similar to flux reconstruction. This way the component averages can be further leveraged to improve sensor temperature approximations as δ perturbs the sensor with respect to the elements.

By varying the noise level, sensor position, and LCM discretization, the mean errors in TCR inference can be generated as shown in Fig. 4. Observe that LCM 4 performs roughly the same as the 1D-FEM in terms of sensor placement, likely due to its similar discretization; without more elements in the Acusil layer, the temperature approximation of the sensor further from the interface is inadequate, producing a poorly inferred conductance value. However, for finer discretized models (e.g., LCM 5 and so on), the error is much smaller and more uniform, an artifact of the better sensor approximation. However, between these finely discretized models, there is not much improvement; that is, there is no real improvement from putting more than two elements in Acusil, as these two already map the temperature gradient quite well. Finally, in terms of noise level, the errors are similar to 1D-FEM for all models — error increases slightly as the noise level increases, likely due to the biases that get introduced through the noise/denoise workflow.

Similarly, the standard deviations for the error in inferred conductances is shown in Fig. 5. Similar to 1D-FEM, the standard deviation increases as a function of noise level, with moderate noise having higher deviation than no noise, and high noise more than both. However, contrarily to the 1D-FEM and LCM 4, the finely discretized models have more uniform standard deviations. This is likely due to their overall low error, which maintains the noise level as a dominant error mode as opposed to the sensor placement in the coarser models.

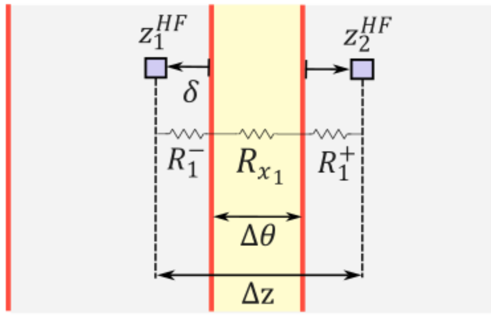


Figure 3: Demonstrative example of an arbitrary sensor displaced from its respective interface position, showing the varying effective resistance with increasing δ.

1. Discuss sensor placement and how it is done for this specific system (i.e. moving indices in FEM)
2. Tradeoff for C vs δ vs SNR (however, this might be best placed in the appendix with the main takeaways in the main document)

5.2.3 Remarks

With the previous analysis of the 1D-FEM and LCMs, the best choice in training for PIROM is clearly LCM 5. In Fig. 4, it is apparent that the finer models have constantly negligible error regardless of sensor position. And while there is no further improvement in error for even finer models (e.g., LCM 6-8), there is a considerable increase in runtime effort. LCM 5, which only has 5 elements, remains computationally tractable while still displaying the benefit of additional elements in Acusil. Furthermore, LCM 5 displays steady behavior across noise levels, and only sees a comparable rise in standard deviation with moderate noise compared to the other models.

Finally, the moderate noise case is the best case to train the PIROM on, since it permits for the analysis of the effects of noise on the system while maintaining a noise intensity similar to that of real sensors. Additionally, it shows the effects of noise while still being physical; the increase in mean and deviation error rise slightly as expected, but they are still tractable for a solver. While the models were able to handle the high noise case, the intensity is quite a bit more extreme than is reasonable and is meant as more of a demonstrative example of the limitations of the model and the trend of the models with gradually increasing noise.

1. What model won? E.g. Model A with Noise B performed the best
2. All else should perhaps be placed in the appendix, with this section only highlighting what model we will compare the PIROM to (to reduce total number of combinations and overall noise in the paper)

5.2.4 Model Performance on Transient-Dominated Timescale

For the experiments conducted in the previous sections, the timescale was fixed and sufficiently long; in other words, the simulation time in which the models ran for was extensive enough to

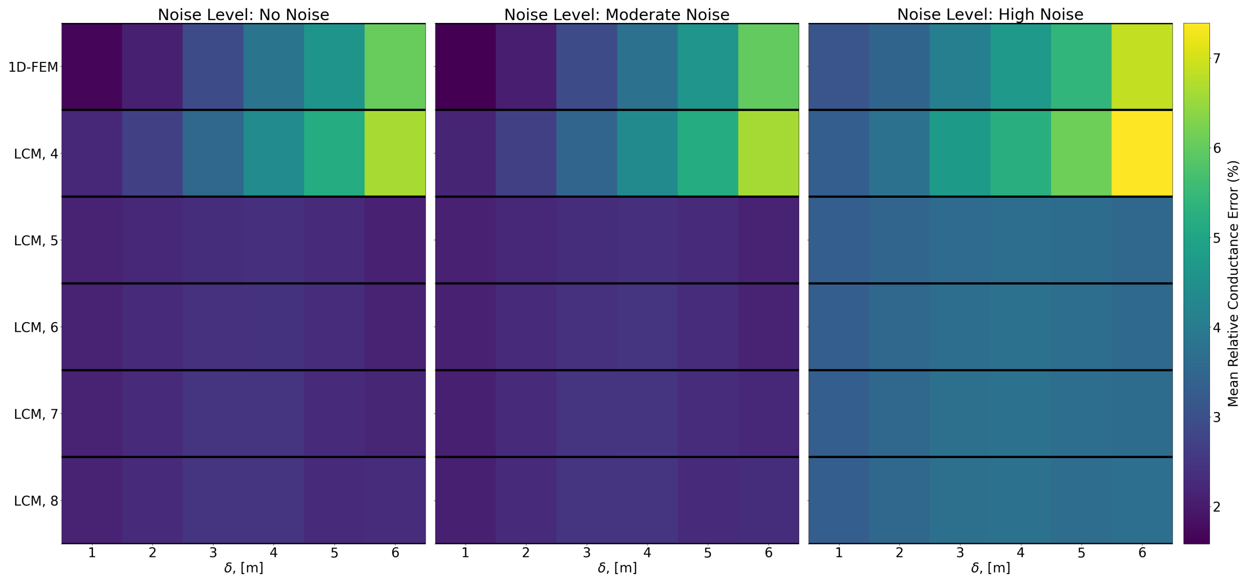


Figure 4: Mean relative error of the inferred conductance values across a range of experiments and model selections.

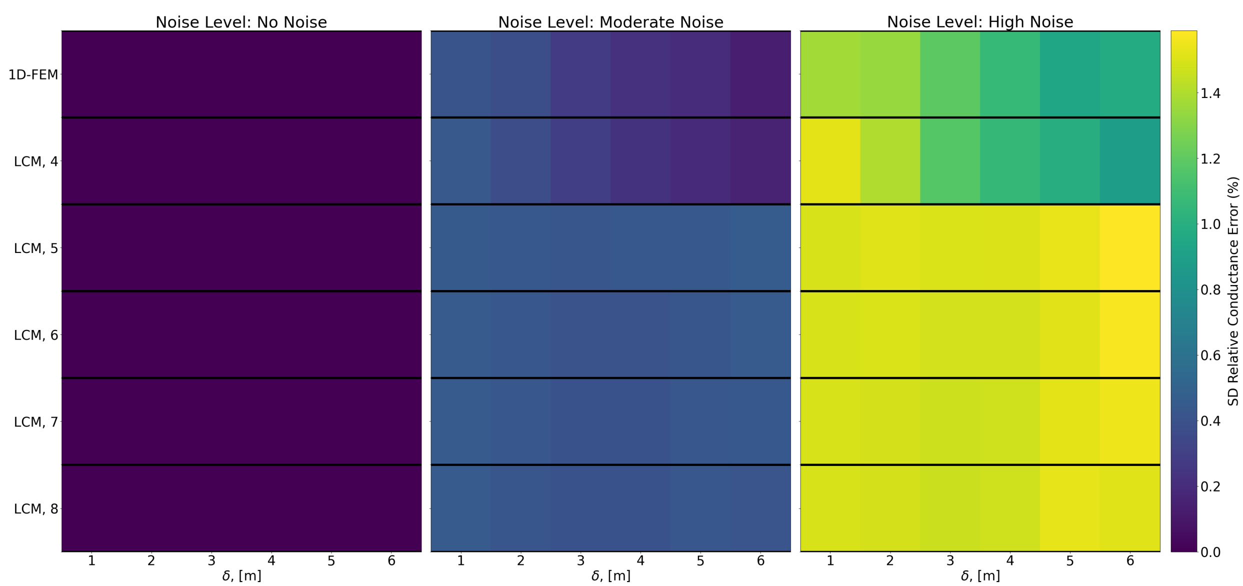


Figure 5: The standard deviation of the relative error for the inferred conductance values from a range of experiments and model selections.

ensure the heat applied on the boundary condition was able to propagate through the entirety of the system. Through this, the inferred conductance inherently satisfies the steady-state effects, of whose error would dominate the cost function used to train. However, such a result brings to question the transient effects. When the timescale is short enough where the initial transients dominates — that is, times where heat has not been able to fully penetrate the system — the inferred conductances should have higher error as the system tries to compensate for effects it can not fully observe yet.

Consider Fig. 6, which shows the error in each inferred conductance as the timescale of the simulation is altered. Observe that there is a certain simulation timescale in which the relative error drops below an adequately low threshold, which perfectly coincides with the effective characteristic time of each of the gaps. In other words, the error drops off the moment where the flux reaches the sensor located immediately after each of the respective gaps. This finding is meaningful; the time needed to infer the conductance of a gap is only the time it takes for heat to travel to the sensors on either side of the gap. Therefore, the framework does not directly necessitate steady-state, but merely must be able to observe the temperatures on either side of the gap to read the physics across it.

For the remainder of the study, the timescale will remain adequately large to ensure sufficient heat penetration through the system; as the PIROM will mainly focus on improved tracking of sensor readings, its ability to improve transient and steady-state effects is non-trivial. However, through these results, the framework is proven capable of inferring material properties within significantly reduces timescales — even ones which are still dominated by transients — as long as there is observation on either side of the gap in question.

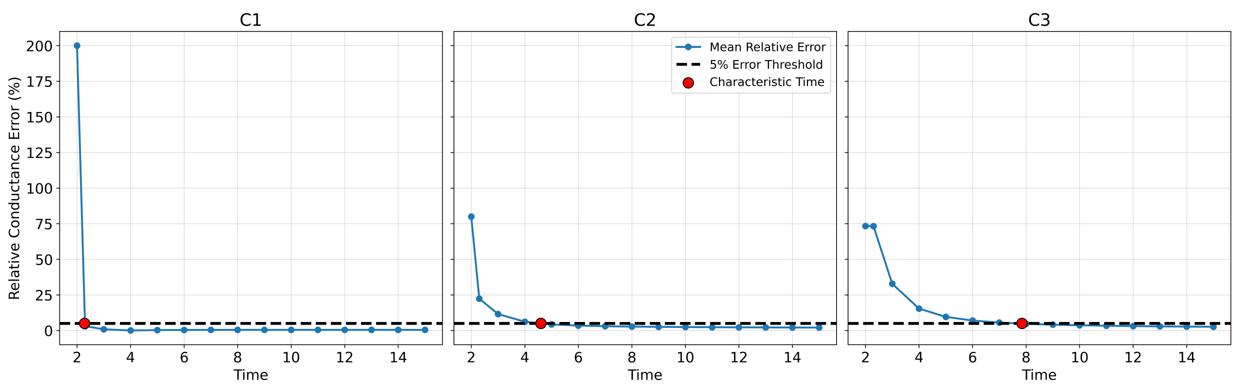


Figure 6: Evolution of error in inferred conductance as a function of simulation timescale.

5.3 PIROM Results

5.3.1 PIROM Optimization

1. Hidden dynamics
2. TCR
3. Training costs
4. Generalization (DONE BELOW)
5. HS Graph??

5.3.2 BC Generalization

For the nominal experiment, the boundary condition (i.e., the flux applied on the edge of the TPS) is constant throughout the simulation. However, in a real experiment, this is not always true; there are small fluctuations in the boundary conditions that the PIROM must continue to track. To do so, a sinusoidal contribution will be added to the forcing applied to the TPS.

Consider the flux component applied to the edge of the TPS given by

$$q = \sigma \cdot [1 + \sin(\omega t)] \tag{55}$$

where σ and ω represent the magnitude and periodic frequency of the forcing respectively. Previously, ω is zero, signifying a constant signal. However, by randomly sampling ω, a periodic element can perturb the signal while still maintaining grounding in the same general vicinity of the nominal forcing.

The results of such perturbation are shown in Fig. 7, where the NRMSE of PIROM is quite significantly lower than that of both the LCM and FEM. This is an artifact of the improved tracking of the PIROM through the learned hidden states, which as expected, extend to other boundary conditions without further training. This is advantageous as only the nominal boundary conditions (i.e., constant flux) must be explicitly learned by PIROM, and the learned hidden states can be implicitly applied to any set of boundary conditions and show major improvement. A further advantage of this is warm starting; even if the error is not adequately low, it provides a much better starting point for training than cold starting would, significantly decreasing runtime for a subsequent training instance.

5.3.3 Material Generalization

Another important application to test is material generalization. At high temperatures, material properties may carry more uncertainty and in practice, may not exactly match the properties that the PIROM was trained on. Thus, PIROM having the capability to extend to slightly perturbed properties is imperative in ensuring its improved tracking retains accuracy in high temperature supersonic environments.

To model this, the material properties can be formulated as

$$m^{(p)}(\bar{T}) = m^{(n)}(\bar{T}) + 0.5\Delta m^{(n)} \left[\alpha_0 \bar{T} + \alpha_1 \sin(\alpha_2 \pi \bar{T})\right] \tag{56}$$

where $m^{(p)}$ and $m^{(n)}$ are the perturbed and nominal material properties respectively, $\bar{T}$ is the normalized temperature given by $\bar{T} = \frac{T - T_{\min}}{T_{\max} - T_{\min}}$, and $\xi = [\alpha_0, \alpha_1, \alpha_2]$ are the random parameters sampled to perturb the material properties. Such a model succeeds in keeping material properties true at low temperatures, but allowing larger perturbations as temperature continues to climb. The generated perturbations are illustrated in Fig. 5.3.3, where sufficient deviation from the nominal properties is demonstrated.

5.3.4 TCR Generalization

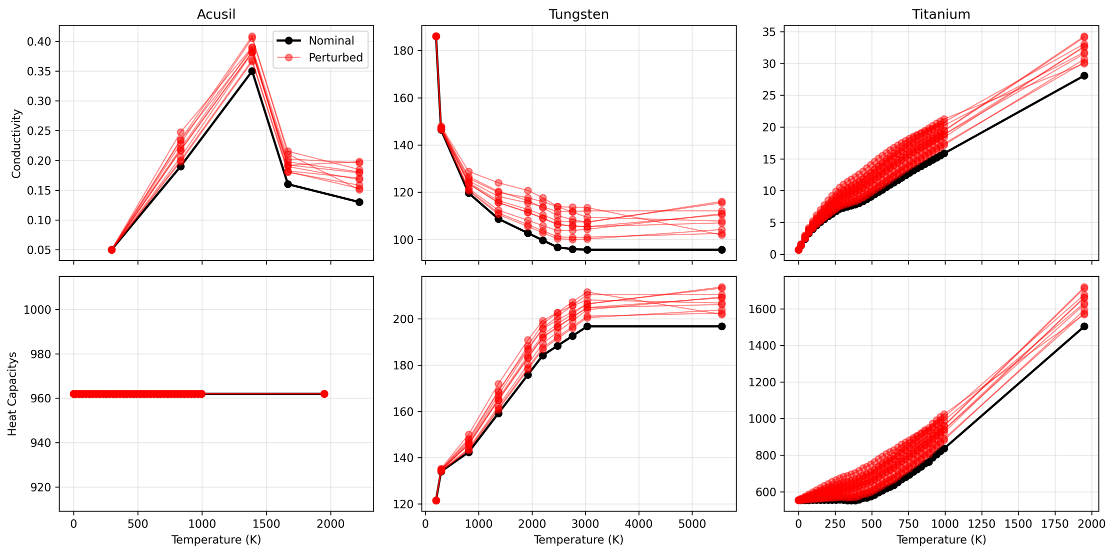
5.3.5 Thickness Generalization

5.3.6 Compounded Generalization

5.3.7 Extreme Thickness Generalization

6 Concluding Remarks

The main conclusion of this work are as follows,



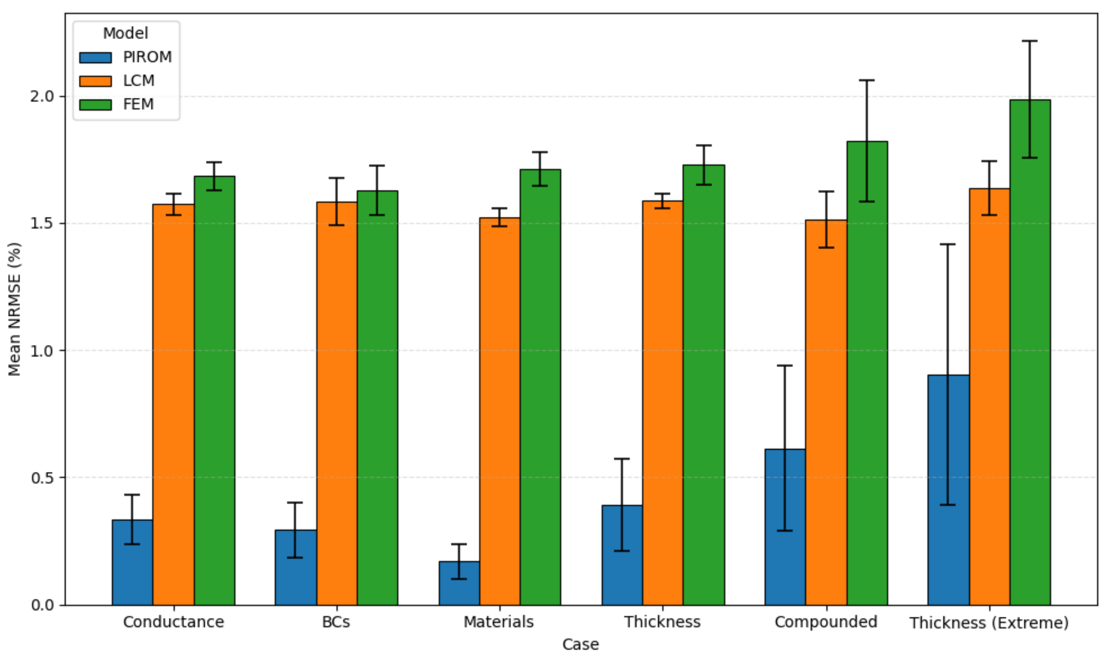


Figure 7: Mean NRMSE of sensor tracking for each of the generalization cases.

1. A one-way physics-data PIROM design is found to successully infer the unknown TCRs and hidden dynamics simulatneously. During inference, X and Y orders of magnitude in *training efficiency* are obtained relative to the FOM and 1D-FEM forward models, while X and Y orders of magnitude in *evaluation efficiency*

   X and Y orders of magnitude in *training efficiency* are obtained relati

   successfully infers the unknown TCRs and hidden dynamics simultaneously, offering X and Y orders of magnitude in *training efficiency*, as well as X and Y

   relative to the FOM and 1D-FEM based model, while X and Y orders of magnitude in *evlauation efficiency*

   and X and Y orders of magnitude in *evaluation efficiency*

   TCRs from noisy thermocouple

   framework successfully learns a ROM for the prediction of the temperature dynamics of the TPS, while simultaneously inferring the unknown contact conductances with errors of less than X% on noisy data with SNR of up to Y dB.

2. The PIROM outperforms the state-of-the-art data-driven, NN-based SciML, and traditional FEM modeling approaches for inferring contact conductances, as well as for predicting the temperature dynamics for OOD configurations.

3. Model bias is found to significantly affect the

A Appendix

This section provides the technical details for the main text, including the DG-FEM formulation of the FOM, as well as the coarse-graining procedure to derive the LCM with interface temperature jump dynamics.

A.1 Discontinuous Galerkin Finite-Element Formulation

Domain Discretization Consider the partition of the TPS spatial domain into N_C continuous components $\{\Omega_i\}_{i=1}^{N_C}$, where each component has its own temperature-dependent material properties. The domain of each component Ω_i is further divided into two sub-domains: (i) $\Omega_i^c \subset \Omega_i$ consisting of a thin layer of material that *is in direct contact with neighboring components across an imperfect interface*, and (ii) $\Omega_i^b \subset \Omega_i$ consisting of the *remaining bulk material that does not have boundaries in direct contact with neighboring components*. Note that one component may have multiple imperfect-contact interfaces, and thus may have multiple contact sub-domains Ω_i^c. For example, in Fig. 1 the domain Ω_2 has two neighboring components with imperfect contacts over interfaces Γ_1 and Γ_3, and thus has two contact sub-domains $\Omega_2^{c,1}$ and $\Omega_2^{c,3}$, and one bulk sub-domain Ω_2^b; clearly, $\Omega_2 = \Omega_2^{c,1} \cup \Omega_2^{c,3} \cup \Omega_2^b$ and $\Omega_2^{c,1} \cap \Omega_2^{c,3} = \emptyset$.

Variational Formulation Choosing appropriate basis functions $v(x)$ and $w(x)$, the variational formulation of the governing equations is obtained element-wise for each component in the TPS. To aid in the subsequent model derivations, denote $\mathcal{N}_i$ as the collection of neighboring elements to E_i, which are split into,

$$\mathcal{N}_i = \mathcal{N}_i^b \cup \mathcal{N}_i^c, \quad i = 1, \ldots, M \tag{57}$$

where $\mathcal{N}_i^b = \mathcal{N}_i^u \cup \mathcal{N}_i^w$ includes the indices of neighboring elements on the Ω_i^b and Ω_i^c sub-domains *across perfect interfaces*. An additional set of neighboring elements *across imperfect interfaces* $\mathcal{N}_i^c$ is introduced. Both sets are formally defined as,

$$\mathcal{N}_i^b = \{j \in \{1, \ldots, M\} \mid E_j \cap E_i = e_{ij} \neq \emptyset, e_{ij} \subseteq \Omega\}\,, \tag{58a}$$

$$\mathcal{N}_i^c = \{j \in \{1, \ldots, M\} \mid \partial E_i \cap \partial E_j \cap \Gamma_r \neq \emptyset\}\,, \quad r = 1, 2, \ldots, N_\Gamma \tag{58b}$$

where $\{\Gamma_r\}_{r=1}^{N_\Gamma}$ correspond to the collection of imperfect interfaces. Under this separation, the Interior Penalty Galerkin (IPG) scheme yields the following bi-linear form for the variational formulation of the governing equations [Cohen2018],

$$\sum_{i=1}^{M} [a_{1,i}(v,w) + a_{2,i}(v,w) + a_{3,i}(v,w)] = \sum_{i=1}^{M} L_i(v) \tag{59}$$

where ϵ is a user-specified parameter, and,

$$a_{1,i}(v,w) = \int_{E_i} \left(\rho_i c_{p,i} v \frac{\partial w}{\partial t} + \mathbf{k}_i \nabla v \cdot \nabla w\right) dE_i \tag{60a}$$

$$\begin{aligned} a_{2,i}(v,w) = &- \sum_{j \in \mathcal{N}_i^b \cup \{T\}} \int_{e_{ij}} \{\mathbf{k}_i \nabla v \cdot \mathbf{n}_i\}\,[w]\, de_{ij} + \epsilon \sum_{j \in \mathcal{N}_i^b \cup \{T\}} \int_{e_{ij}} \{\mathbf{k}_i \nabla w \cdot \mathbf{n}_i\}\,[v]\, de_{ij} \\ &+ \sigma_b \sum_{j \in \mathcal{N}_i^b \cup \{T\}} \int_{e_{ij}} [v]\,[w]\, de_{ij} \end{aligned} \tag{60b}$$

$$\begin{aligned} a_{3,i}(v,w) = &- \sum_{j \in \mathcal{N}_i^c} \int_{\Gamma_{ij}} \{\mathbf{k}_i \nabla v \cdot \mathbf{n}_i\}\,[w]\, d\Gamma + \epsilon \sum_{j \in \mathcal{N}_i^c} \int_{\Gamma_{ij}} \{\mathbf{k}_i \nabla w \cdot \mathbf{n}_i\}\,[v]\, d\Gamma \\ &+ \sigma_c \sum_{j \in \mathcal{N}_i^c} \int_{\Gamma_{ij}} [v]\,[w]\, d\Gamma \end{aligned} \tag{60c}$$

$$L_i(v) = \epsilon \sum_{j \in \mathcal{N}_i^b \cup \{T\}} \int_{e_{ij}} (\mathbf{k}_i \nabla w \cdot \mathbf{n}_i)\, T_b de_{ij} + \int_{e_{iq}} v q_b de_{iq} + \sigma \int_{e_{iT}} v T_b de_{iT} \tag{60d}$$

where $a_{1,i}$, $a_{2,i}$, $a_{3,i}$, and $L_i(v)$ amount to heat conduction due to bulk energy transfer, between-element perfect interfaces e_{ij}, between-element imperfect interfaces Γ_{ij}, and boundary conditions, respectively. In the bi-linear forms, the notations of [] and {} are respectively the jumps and averages at an interface e_{ij} shared by two neighboring elements E_i and E_j, defined as,

$$[u] = u\Big|_{E_i} - u\Big|_{E_j}, \quad \{u\} = \frac{1}{2}\left(u\Big|_{E_i} + u\Big|_{E_j}\right), \quad x \in e_{ij} = \partial E_i \cap \partial E_j \tag{61}$$

where the same holds for an imperfect interface Γ_{ij}, and where σ_b, σ_c are known user-specified bulk and interface penalty terms whose value depends on the element size. Depending on the choice of ϵ, the bi-linear form may correspond to symmetric IPG ($\epsilon = -1$), non-symmetric IPG ($\epsilon = 1$), or incomplete IPG ($\epsilon = 0$) schemes. All these schemes are consistent with the original PDE and have a similar convergence rate with respect to mesh size.

Galerkin Projections Next, the DG-FEM is written in an element-wise form. For the i-th element, use a set of P basis functions, such as polynomials, to represent the temperature field as in Eq. (2). Using the same test functions as trial functions, the DG-FEM dynamics for the i-th element is obtained by evaluating the element-wise bi-linear forms,

$$a_{1,i}(\phi_k^i, T) + a_{2,i}(\phi_k^i, T) + a_{3,i}(\phi_k^i, T) = L_i(\phi_k^i), \quad k = 1, \ldots, P \quad i = 1, 2, \ldots, M \tag{62}$$

where ϕ_k^i is the k-th basis function for the i-th element. Evaluating for all basis functions, while leveraging the splitting of DG-FEM states into bulk and contact elements, the following system of coupled ODEs is obtained,

- **Bulk elements**: for $i = 1, \ldots, B$,

$$\mathbf{A}_i(\mathbf{u}_i)\dot{\mathbf{u}}_i = \underbrace{\left[\mathbf{B}_i(\mathbf{u}_i)\mathbf{u}_i + \sum_{j\in\mathcal{N}_i^b} \mathbf{B}_{ij}^i \mathbf{u}_i + \sum_{j\in\mathcal{N}_i^u} \mathbf{B}_{ij}^j \mathbf{u}_j\right]}_{\text{bulk dynamics on } \Omega_i^b} + \underbrace{\left[\sum_{j\in\mathcal{N}_i^w} \mathbf{B}_{ij}^j \mathbf{w}_j\right]}_{\text{coupling to dynamics on } \Omega_i^c} + \underbrace{\mathbf{f}_i^{(b)}(t)}_{\text{heating BC}} \tag{63}$$

- **Imperfect interface elements**: for $i = 1, \ldots, K$,

$$\begin{aligned}\mathbf{D}_i(\mathbf{w}_i)\dot{\mathbf{w}}_i = &\underbrace{\left[\sum_{j\in\mathcal{N}_i^u} \mathbf{B}_{ij}^j \mathbf{u}_j\right]}_{\text{coupling to dynamics on } \Omega_i^u} + \underbrace{\left[\mathbf{B}_i(\mathbf{w}_i)\mathbf{w}_i + \sum_{j\in\mathcal{N}_i^b} \mathbf{B}_{ij}^i \mathbf{w}_i + \sum_{j\in\mathcal{N}_i^w} \mathbf{B}_{ij}^j \mathbf{w}_j\right]}_{\text{bulk dynamics on } \Omega_i^c} \\ &+ \underbrace{\left[\sum_{j\in\mathcal{N}_i^c} \left(\mathbf{K}_{ij}^i \mathbf{w}_i + \mathbf{K}_{ij}^j \mathbf{w}_j\right)\right]}_{\text{imperfect contact dynamics due to } \Gamma_r} + \underbrace{\mathbf{f}_i^{(c)}(t)}_{\text{heating BC}}\end{aligned} \tag{64}$$

where $M = K + B$. For $k, l = 1, \ldots, P$, the matrix elements are given by,

$$[\mathbf{A}_i]_{kl} = \int_{E_i} \rho_i c_{p,i} \phi_k^i \phi_l^i dE_i \tag{65a}$$

$$[\mathbf{B}_i]_{kl} = \int_{E_i} \mathbf{k}_i \nabla\phi_k^i \cdot \nabla\phi_l^i dE_i \tag{65b}$$

$$\left[\mathbf{B}_{ij}^i\right]_{kl} = \begin{cases} -\int_{e_{ij}} \left\{\mathbf{k}\nabla\phi_k^i \cdot \mathbf{n}_i\right\} \phi_l^i + \sigma_b \left[\phi_k^i\right] \phi_l^i d\Gamma, & j \in \mathcal{N}_i^u \cup \{T\} \\ 0, & \text{otherwise} \end{cases} \tag{65c}$$

$$\left[\mathbf{B}_{ij}^j\right]_{kl} = \begin{cases} \int_{e_{ij}} \left\{\mathbf{k}\nabla\phi_k^i \cdot \mathbf{n}\right\} \phi_l^j + \sigma_b \left[\phi_k^i\right] \phi_l^j d\Gamma, & j \in \mathcal{N}_i^u \cup \{T\} \\ 0, & \text{otherwise} \end{cases} \tag{65d}$$

$$\left[\mathbf{K}_{ij}^i\right]_{kl} = \begin{cases} -\int_{\Gamma_{ir}} \left\{\mathbf{k}\nabla\phi_k^i \cdot \mathbf{n}_i\right\} \phi_l^i + \sigma_c \left[\phi_k^i\right] \phi_l^i d\Gamma, & j \in \mathcal{N}_i^c \\ 0, & \text{otherwise} \end{cases} \tag{65e}$$

$$\left[\mathbf{K}_{ij}^j\right]_{kl} = \begin{cases} \int_{\Gamma_{ir}} \left\{\mathbf{k}\nabla\phi_k^i \cdot \mathbf{n}\right\} \phi_l^j + \sigma_c \left[\phi_k^i\right] \phi_l^j d\Gamma, & j \in \mathcal{N}_i^c \\ 0, & \text{otherwise} \end{cases} \tag{65f}$$

$$\left[\mathbf{f}_i^{(b)}\right]_k = \int_{e_{iq}} \phi_k^i q_b de + \sigma \int_{e_{iT}} \phi_k^i T_b de \tag{65g}$$

$$\left[\mathbf{f}_i^{(c)}\right]_k = \int_{\Gamma_{ir}} \phi_k^i q_c de + \sigma \int_{\Gamma_{iT}} \phi_k^i T_c de \tag{65h}$$

where the matrices $\mathbf{A}_i$, $\mathbf{B}_i$, $\mathbf{B}_{ij}^i$, and $\mathbf{B}_{ij}^j$ depend on ρ, c_p, and $\mathbf{k}$, respectively, which are non-linear functions of temperature and thus become the main source of non-linearities in the problem. The TCRs are assumed constant in this work, and thus the stiffness matrices due to imperfect contacts $\mathbf{K}_{ij}^i$ and $\mathbf{K}_{ij}^j$ are temperature-independent. Moreover, since the basis functions are orthogonal ρc_p is constant within an element, and $\mathbf{A}_i$ becomes a diagonal matrix; otherwise, $\mathbf{A}_i$ is symmetric and positive definite, as $\rho c_p > 0$.

For compactness, the element-wise ODEs in Eqs. (63) to (64) are stacked together to obtain the following system of ODEs for all bulk and interfacing elements in the TPS,

$$\boldsymbol{\mathcal{A}}(\mathbf{x})\dot{\mathbf{x}} = \boldsymbol{\mathcal{B}}(\mathbf{x})\mathbf{x} + \boldsymbol{\mathcal{F}}(t) \tag{66}$$

where $\mathbf{x} = [\mathbf{u}, \mathbf{w}]^\top \in \mathbb{R}^{MP}$ are the states, and the matrices are given by,

$$\boldsymbol{\mathcal{A}}(\mathbf{x}) = \begin{bmatrix} \mathbf{A}(\mathbf{u}) & \mathbf{0} \\ \mathbf{0} & \mathbf{D}(\mathbf{w}) \end{bmatrix}, \; \boldsymbol{\mathcal{B}}(\mathbf{x}) = \begin{bmatrix} \mathbf{B}(\mathbf{u}) & \mathbf{C}(\mathbf{u}) \\ \mathbf{E}(\mathbf{u}) & \mathbf{F}(\mathbf{w}) \end{bmatrix}, \; \boldsymbol{\mathcal{F}}(t) = \begin{bmatrix} \mathbf{f}^{(b)}(t) \\ \mathbf{f}^{(c)}(t) \end{bmatrix} \tag{67}$$

where $M = B + K$ and $\mathbf{B} \in \mathbb{R}^{BP\times BP}$, $\mathbf{C} \in \mathbb{R}^{BP\times KP}$, $\mathbf{E} \in \mathbb{R}^{KP\times BP}$, and $\mathbf{F} \in \mathbb{R}^{KP\times KP}$, with Neumann and Dirichlet boundary conditions assumed to be applied only to the bulk domains. Given that the neighboring elements $\mathcal{N}_i$ of E_i are sub-divided into bulk $\mathcal{N}_i^b = \mathcal{N}_i^u \cup \mathcal{N}_i^w$ and contact $\mathcal{N}_i^c$, the $M \times M$ block structure to the mass $\boldsymbol{\mathcal{A}}$ and stiffness $\boldsymbol{\mathcal{B}}$ matrices is introduced,

$$\mathbf{A}_{ij}(\mathbf{u}) = \begin{cases} \mathbf{A}_i(\mathbf{u}_i) & i = j, \\ \mathbf{0} & \text{otherwise} \end{cases}, \quad i, j = 1, \dots, B \tag{68a}$$

$$\mathbf{D}_{ij}(\mathbf{w}) = \begin{cases} \mathbf{D}_i(\mathbf{w}_i) & i = j, \\ \mathbf{0} & \text{otherwise} \end{cases}, \quad i, j = 1, \dots, K \tag{68b}$$

$$\mathbf{B}_{ij}(\mathbf{u}) = \begin{cases} \mathbf{B}_i(\mathbf{u}) + \sum_{l\in\mathcal{N}_i^b} \mathbf{B}_{il}^i(\mathbf{u}) & i = j, \\ \mathbf{B}_{ij}^j(\mathbf{u}) & i \neq j \text{ and } j \in \mathcal{N}_i^b \\ \mathbf{0} & \text{otherwise} \end{cases} \quad i, j = 1, \dots, B \tag{68c}$$

$$\mathbf{C}_{ij}(\mathbf{u}) = \begin{cases} \mathbf{B}_{ij}^j & j \in \mathcal{N}_i^w \\ \mathbf{0} & \text{otherwise} \end{cases} \quad i = 1, \dots, B, \; j = 1, \dots, K \tag{68d}$$

$$\mathbf{E}_{ij}(\mathbf{u}) = \begin{cases} \mathbf{B}_{ij}^j & j \in \mathcal{N}_i^u \\ \mathbf{0} & \text{otherwises} \end{cases} \quad i = 1, \dots, K, \; j = 1, 2, \dots, B \tag{68e}$$

$$\mathbf{F}_{ij}(\mathbf{w}) = \begin{cases} \mathbf{B}_i(\mathbf{w}_i) + \sum_{l\in\mathcal{N}_i^b} \mathbf{B}_{il}^i + \sum_{l\in\mathcal{N}_i^c} \mathbf{K}_{il}^i & i = j \\ \mathbf{B}_{ij}^j & i \neq j \text{ and } j \in \mathcal{N}_i^w \\ \mathbf{K}_{ij}^j & i \neq j \text{ and } j \in \mathcal{N}_i^c \\ \mathbf{0} & \text{otherwise} \end{cases} \quad i, j = 1, \dots, K \tag{68f}$$

A.2 Reduced-Physics Model

A *lumped capacitance model* (LCM) under imperfect TPS layer contacts is formulated in this section as the physics-based backbone of the PIROM formulation. The LCM is a classical physics-based low-order model for predicting the temporal variation of average temperatures in multiple interconnected components [INCROPERA]. The LCM is derived at the component level from a point of view of energy conservation, where the following assumptions are employed: (i) bulk Ω_i^b and interface Ω_i^c components have uniform temperatures $\bar{u}_i$, respectively, which are collected into an average variable $\bar{\mathbf{x}}_i = [\bar{u}_i, \bar{w}_i]^\top$; (ii) between neighboring i and j, the heat flux is given by

$q_{ij} = g_{ij}(\bar{\mathbf{x}})(\bar{u}_i - \bar{u}_j)$ for $i, j \in \mathcal{N}_i^b$ through a *perfect*, and $q_{ij} = g_{x,ij}(\bar{w}_i - \bar{w}_j)$ for $i, j \in \mathcal{N}_i^c$ through an *imperfect* contact.

... in progress

A.3 Coarse-Graining Procedure

This section provides the details of coarse-graining for the TPS problem, the magnitude analysis of residual dynamics after coarse-graining, and an illustrative 2D example demonstrating the derivation of the LCM from the DG-FEM FOM.

A.3.1 Detailed Derivation of Coarse-Graining

This section supplements the details pertaining to the coarse-graining procedure presented in Sec. X. First, by definition,

$$\begin{aligned}
\mathbf{r}^{(1)}(\bar{\mathbf{x}}, t) &= \mathcal{P}\left[\mathbf{X}^\dagger \mathbf{r}(\mathbf{x}, t)\right] && \text{(69a)}\\
&= \mathcal{P}\left[\mathbf{X}^\dagger \boldsymbol{\mathcal{A}}(\mathbf{x})^{-1}\boldsymbol{\mathcal{B}}(\mathbf{x})\mathbf{x} + \mathbf{X}^\dagger \boldsymbol{\mathcal{A}}(\mathbf{x})^{-1}\boldsymbol{\mathcal{F}}(t)\right] && \text{(69b)}\\
&= \mathcal{P}\left[\mathbf{X}^\dagger \boldsymbol{\mathcal{A}}(\mathbf{x})^{-1}\mathbf{B}(\mathbf{x})\mathbf{x}\right] + \mathcal{P}\left[\mathbf{X}^\dagger \boldsymbol{\mathcal{A}}(\mathbf{x})^{-1}\boldsymbol{\mathcal{F}}(t)\right] && \text{(69c)}\\
&= \mathbf{X}^\dagger \boldsymbol{\mathcal{A}}(\mathbf{P}\mathbf{x})^{-1}\mathbf{B}(\mathbf{P}\mathbf{x})\mathbf{P}\mathbf{x} + \mathbf{X}^\dagger \boldsymbol{\mathcal{A}}(\mathbf{P}\mathbf{x})^{-1}\mathbf{P}\boldsymbol{\mathcal{F}}(t) && \text{(69d)}\\
&= \underbrace{[\mathbf{X}^\dagger \mathbf{A}(\mathbf{X}\bar{\mathbf{x}})^{-1}\mathbf{X}]}_{\#1}\underbrace{[\mathbf{X}^\dagger \mathbf{B}(\mathbf{X}\bar{\mathbf{x}})\mathbf{X}]}_{\#2}\bar{\mathbf{x}} + \mathbf{X}^\dagger \boldsymbol{\mathcal{A}}(\mathbf{X}\bar{\mathbf{x}})^{-1}\mathbf{X}\underbrace{[\mathbf{X}^\dagger \boldsymbol{\mathcal{F}}(t)]}_{\#3} && \text{(69e)}
\end{aligned}$$

where the terms #1, #2, and #3 are the coarse-grained representation of the inverted heat capacitance, heat conduction, and heat load matrices. Following the coarse-graining procedure developed in our previous work Ref. [VargasVenegas20206], these terms are treated one by one to demonstrate the equivalence between the LCM matrices and coarsened DG-FEM matrices.

Term #1: The capacitance matrix $\boldsymbol{\mathcal{A}}$ is block diagonal, and composed of the bulk $\mathbf{A}$ and interface $\mathbf{D}$ capacitance matrices. These bulk and contact capacitance matrices are themselves diagonal, with their i-th diagonal blocks as $\mathbf{A}_i \in \mathbb{R}^{P\times P}$ and $\mathbf{D}_i \in \mathbb{R}^{P\times P}$, which become diagonal if the material properties are constant within the element. Take an element $E_i \in \Omega_j$ and denote $\overline{\rho c_p}_j$ as the value of ρc_p evaluated at the average $\bar{u}_j$ so that $\mathbf{A}_i = \overline{\rho c_p}_j |B_i| \mathbf{I}$ and $\mathbf{D}_i = \overline{\rho c_p}_j |W_i| \mathbf{I}$ with $|E_i|$ as the element size. The (k,l)-th element of the coarse-grained *bulk* and *contact* capacitance matrices are computed next.

Coarse-Grained Capacitance Matrices

The coarse-grained *bulk* capacitance matrix is computed as,

$$\begin{aligned}
\left[\boldsymbol{\Phi}^\dagger \mathbf{A}(\boldsymbol{\Phi}\bar{\mathbf{u}})^{-1}\boldsymbol{\Phi}\right]_{kl} &= \sum_{i=1}^{B}\sum_{j=1}^{B} \boldsymbol{\varphi}_i^{k\dagger}\mathbf{A}_{ij}(\boldsymbol{\Phi}\bar{\mathbf{u}})^{-1}\boldsymbol{\varphi}_j^l && \text{(70a)}\\
&= \sum_{i=1}^{B}\sum_{j=1}^{B} \boldsymbol{\varphi}_i^{k\dagger}\left[\left(\overline{\rho c_p}_k |B_i|\right)^{-1}\mathbf{I}\delta_{ij}\right]\boldsymbol{\varphi}_j^l && \text{(70b)}\\
&= \sum_{i=1}^{B} \boldsymbol{\varphi}_i^{k\dagger}\left(\overline{\rho c_p}_k |B_i|\right)^{-1}\boldsymbol{\varphi}_i^l && \text{(70c)}
\end{aligned}$$

$$= \sum_{i\in\mathcal{V}_k} \boldsymbol{\varphi}_i^{k\dagger} \left(\overline{\rho c_p}_k \, |B_i|\right)^{-1} \boldsymbol{\varphi}_i^l \tag{70d}$$

$$= \sum_{i\in\mathcal{V}_k} \frac{|B_i|}{|\Omega_k^b|} \boldsymbol{\varphi}_i^{k\top} \left(\overline{\rho c_p}_k \, |B_i|\right)^{-1} \boldsymbol{\varphi}_i^k \tag{70e}$$

$$= \sum_{i\in\mathcal{V}_k} \frac{1}{\Omega_k^b} \boldsymbol{\varphi}_i^{k\top} \left(\overline{\rho c_p}_k\right)^{-1} \boldsymbol{\varphi}_i^k \tag{70f}$$

$$= \sum_{i\in\mathcal{V}_k} \left(\overline{\rho c_p}_k \left|\Omega_k^b\right|\right)^{-1} \tag{70g}$$

$$= |\mathcal{V}_k| \left(\overline{\rho c_p}_k \left|\Omega_k^b\right|\right)^{-1} \tag{70h}$$

Similarly, the coarse-grained *contact* capacitance matrix is computed as,

$$\left[\boldsymbol{\Pi}^\dagger \mathbf{D} \left(\boldsymbol{\Pi}\bar{\mathbf{w}}\right)^{-1} \boldsymbol{\Pi}\right]_{kl} = \sum_{i=1}^{K}\sum_{j=1}^{K} \boldsymbol{\psi}_i^{k\dagger} \mathbf{D}_{ij} \left(\boldsymbol{\Pi}\bar{\mathbf{w}}\right)^{-1} \boldsymbol{\psi}_j^l \tag{71a}$$

$$= \sum_{i=1}^{K}\sum_{j=1}^{K} \boldsymbol{\psi}_i^{k\dagger} \left[\left(\overline{\rho c_p}_k \, |W_i|\right)^{-1} \mathbf{I}\delta_{ij}\right] \boldsymbol{\psi}_j^l \tag{71b}$$

$$= \sum_{i=1}^{K} \boldsymbol{\psi}_i^{k\dagger} \left(\overline{\rho c_p}_k \, |W_i|\right)^{-1} \boldsymbol{\psi}_i^l \tag{71c}$$

$$= \sum_{i\in\mathcal{F}_k} \frac{|W_i|}{|\Omega_k^c|} \boldsymbol{\psi}_i^{k\top} \left(\overline{\rho c_p}_k \, |W_i|\right)^{-1} \boldsymbol{\psi}_i^k \tag{71d}$$

$$= \sum_{i\in\mathcal{F}_k} \frac{1}{|\Omega_k^c|} \boldsymbol{\psi}_i^{k\top} \left(\overline{\rho c_p}_k\right)^{-1} \boldsymbol{\psi}_i^k \tag{71e}$$

$$= \sum_{i\in\mathcal{F}_k} \left(\overline{\rho c_p}_k \, |\Omega_k^c|\right)^{-1} \tag{71f}$$

$$= |\mathcal{F}_k| \left(\overline{\rho c_p}_k \, |\Omega_k^c|\right)^{-1} \tag{71g}$$

Therefore, by introducing the diagonal weight matrices $\boldsymbol{\mathcal{W}}_V \in \mathbb{R}^{N_B\times N_B}$ and $\boldsymbol{\mathcal{W}}_F \in \mathbb{R}^{N_C\times N_C}$ it follows that,

$$\boldsymbol{\Phi}^\dagger \mathbf{A} \left(\boldsymbol{\Phi}\bar{\mathbf{u}}\right)^{-1} \boldsymbol{\Phi}\boldsymbol{\mathcal{W}}_V^{-1} = \bar{\mathbf{A}} \left(\bar{\mathbf{u}}\right)^{-1}, \quad \boldsymbol{\Pi}^\dagger \mathbf{D} \left(\boldsymbol{\Pi}\bar{\mathbf{w}}\right)^{-1} \boldsymbol{\Pi}\boldsymbol{\mathcal{W}}_F^{-1} = \bar{\mathbf{D}} \left(\bar{\mathbf{w}}\right)^{-1} \tag{72}$$

Note that $|\Omega_k^c| \approx 0$ as the volume of the interfacing domains is reduced, and thus $\bar{\mathbf{D}}\left(\bar{\mathbf{w}}\right) \approx \mathbf{0}$ under a *zero-energy storage assumption for the interacing domain*; such assumption is leveraged in subsequent derivations to demonstrate the equivalence between the coarse-grained DG-FEM model and Eq. (22).

Term #2: The thermal conductivity matrix $\boldsymbol{\mathcal{B}}$ has a block structure that is divided into: $\mathbf{B} \in \mathbb{R}^{B\times B}$ capturing bulk heat transfer, $\mathbf{C} \in \mathbb{R}^{B\times K}$ and $\mathbf{E} \in \mathbb{R}^{K\times B}$ capturing the heat transfer coupling between bulk and contact sub-domains, and $\mathbf{F} \in \mathbb{R}^{K\times K}$ capturing the bulk heat transfer within the contact sub-domains. The (k, l)-th elements of the coarse-grained thermal conductivity matrices are computed next.

Coarse-Grained Thermal Conductivity Matrices

First, the stiffness matrices for the *bulk* and *contact* sub-domains are coarse-grained. The coarse-graining proceeds as follows,

$$\left[\mathbf{\Phi}^{\dagger}\mathbf{B}\left(\mathbf{\Phi}\bar{\mathbf{u}}\right)\mathbf{\Phi}\right]_{kl]} = \sum \tag{73a}$$

$$= \sum \tag{73b}$$

Similarly, coarse-graining the stiffness matrix for bulk heat transfer $\mathbf{F}\left(\mathbf{w}\right)$ within the contact sub-domain proceeds as,

$$\left[\mathbf{\Pi}^{\dagger}\mathbf{F}\left(\mathbf{\Pi}\bar{\mathbf{w}}\right)\mathbf{\Pi}\right]_{kl} = \sum_{i=1}^{K}\sum_{j=1}^{K}\boldsymbol{\psi}_i^{k\dagger}\mathbf{F}_{ij}\left(\mathbf{\Pi}\bar{\mathbf{w}}\right)\boldsymbol{\psi}_j^{l} \tag{74a}$$

$$= \sum_{j\in\mathcal{F}_k}\sum_{j\in\mathcal{F}_l}\frac{|W_i|}{|\Omega_k^c|}\boldsymbol{\psi}_i^{k\top}\mathbf{F}_{ij}\left(\mathbf{\Pi}\bar{\mathbf{w}}\right)\boldsymbol{\psi}_j^{l} \tag{74b}$$

$$= \frac{1}{|\mathcal{F}_k|}\sum_{i\in\mathcal{F}_k}\sum_{j\in\mathcal{F}_l}\left[\mathbf{F}_{ij}\right]_{11} \tag{74c}$$

Using the definition from Eq. (65h) it follows that,

$$\left[\mathbf{F}_{ij}\right]_{11} = \begin{cases} -\sum_{p\in\mathcal{N}_i^b}\sigma_b\left(\bar{\mathbf{u}}\right)|e_{ip}| - \sum_{p\in\mathcal{N}_i^c}\sigma_c\left|\Gamma_{ip}\right|, & i = j \\ \sigma_b(\bar{\mathbf{u}})\left|e_{ij}\right|, & j\in\mathcal{N}_i^w \\ \sigma_c\left|\Gamma_{ij}\right|, & j\in\mathcal{N}_i^c \\ 0, & \text{otherwise} \end{cases} \tag{75}$$

Define the following sets collecting the edges and indices for interfaces with perfect contacts,

$$\mathcal{E}_{kl}^b = \left\{(i,j)\ |\ j\in\mathcal{N}_i^b, i\in\mathcal{F}_k\right\} \tag{76a}$$

$$\bar{\mathcal{N}}_k^b = \left\{l\ |\ \mathcal{E}_{kl}^b \neq \emptyset\right\} \tag{76b}$$

and imperfect contacts,

$$\mathcal{E}_{kl}^c = \{(i,j)\ |\ j\in\mathcal{N}_i^c, i\in\mathcal{F}_k, j\in\mathcal{F}_l\} \tag{77a}$$

$$\bar{\mathcal{N}}_k^c = \{l\ |\ \mathcal{E}_{kl}^c \neq \emptyset\} \tag{77b}$$

so that for $k,l = 1,2,\ldots,N_c$ the sum may be simplified as,

$$\sum_{i\in\mathcal{F}_k}\sum_{j\in\mathcal{F}_l}\left[\mathbf{F}_{ij}\right]_{11} = \begin{cases} -\sum_{p\in\bar{\mathcal{N}}_k^b}\sum_{(i,j)\in\mathcal{E}_{kp}^b}\sigma_b(\bar{\mathbf{u}})\left|e_{ij}\right| - \sum_{p\in\bar{\mathcal{N}}_k^c}\sum_{(i,j)\in\mathcal{E}_{kp}^c}\sigma_c\left|\Gamma_{ij}\right|, & k = l \\ \sum_{(i,j)\in\mathcal{E}_{kl}^w}\sigma_b(\bar{\mathbf{u}})\left|e_{ij}\right|, & l\in\bar{\mathcal{N}}_k^w \\ \sum_{(i,j)\in\mathcal{E}_{kl}^c}\sigma_c\left|\Gamma_{ij}\right|, & l\in\bar{\mathcal{N}}_k^c \\ 0, & \text{otherwise} \end{cases} \tag{78a}$$

$$= \begin{cases} -\sum_{p\in\bar{\mathcal{N}}_k^b}\sigma_b\left(\bar{\mathbf{u}}\right)|e_{kp}| - \sum_{p\in\bar{\mathcal{N}}_k^c}\sigma_c\left|\Gamma_{kp}\right|, & k = l \\ \sigma_b(\bar{\mathbf{u}})\left|e_{kl}\right|, & l\in\bar{\mathcal{N}}_k^w \\ \sigma_c\left|\Gamma_{kl}\right|, & l\in\bar{\mathcal{N}}_k^c \\ 0, & \text{otherwise} \end{cases} \tag{78b}$$

Therefore, by introducing the diagonal weight matrix $\boldsymbol{\mathcal{W}}_V \in \mathbb{R}^{N_c\times N_c}$, the coarse-grained stiffness matrix for the contact sub-domains is written as,

$$\bar{\mathbf{F}}\left(\bar{\mathbf{w}}\right) = \boldsymbol{\mathcal{W}}_F \boldsymbol{\Pi}^\dagger \mathbf{F}\left(\boldsymbol{\Pi}\bar{\mathbf{w}}\right)\boldsymbol{\Pi} \tag{79}$$

The cross-coupling stifness matrices $\mathbf{C} \in \mathbb{R}^{B\times K}$ and $\mathbf{E} \in \mathbb{R}^{K\times B}$ are coarse-grained next. The (k,l)-th element of the coarse-grained $\mathbf{C}$ matrix is computed as,

$$\begin{aligned}
\left[\boldsymbol{\Phi}^\dagger \mathbf{C}\left(\boldsymbol{\Phi}\bar{\mathbf{u}}\right)\boldsymbol{\Pi}\right]_{kl} &= \sum_{i=1}^{B}\sum_{j=1}^{K} \boldsymbol{\varphi}_i^{k\dagger}\mathbf{C}_{ij}\left(\boldsymbol{\Phi}\bar{\mathbf{u}}\right)\boldsymbol{\psi}_j^l &\text{(80a)}\\
&= \sum_{i\in\mathcal{V}_k}\sum_{j\in\mathcal{F}_l} \boldsymbol{\varphi}_i^{k\dagger}\mathbf{C}_{ij}\left(\boldsymbol{\Phi}\bar{\mathbf{u}}\right)\boldsymbol{\psi}_j^l &\text{(80b)}\\
&= \sum_{i\in\mathcal{V}_k}\sum_{j\in\mathcal{F}_l} \frac{|B_i|}{\left|\Omega_k^b\right|}\boldsymbol{\varphi}_i^{k\top}\mathbf{C}_{ij}\left(\boldsymbol{\Phi}\bar{\mathbf{u}}\right)\boldsymbol{\psi}_j^l &\text{(80c)}\\
&= \sum_{i\in\mathcal{V}_k}\sum_{j\in\mathcal{F}_l} \frac{|B_i|}{\left|\Omega_k^b\right|}\left[\mathbf{C}_{ij}\right]_{11} &\text{(80d)}\\
&= \frac{1}{|\mathcal{V}_k|}\sum_{i\in\mathcal{V}_k}\sum_{j\in\mathcal{F}_l}\left[\mathbf{B}_{ij}^j\right]_{11} &\text{(80e)}
\end{aligned}$$

Where it is assumed that all elements within a component have the same size, i.e., $\left|\Omega_k^b\right| = |B_i|\,|\mathcal{V}_k|$, the collection of edges between bulk Ω_k^b and contact Ω_l^c domains is $\mathcal{E}_{kl}^w = \{(i,j)\,|i\in\mathcal{V}_k, j\in\mathcal{N}_i^w, j\in\mathcal{F}_l\}$ so that $|e_{kl}| = \sum_{(i,j)\in\mathcal{E}_{kl}^w}|e_{ij}|$, and where the neighbors of component k are $\bar{\mathcal{N}}_k^w = \{l|\mathcal{E}_{kl}^w \neq \emptyset\}$. By leveraging the definition in Eq. (65h), the sum is simplified as,

$$\sum_{i\in\mathcal{V}_k}\sum_{j\in\mathcal{F}_l}\left[\mathbf{B}_{ij}^j\right]_{11} = \begin{cases}\sum_{(i,j)\in\mathcal{E}_{kl}^w}\sigma_b(\bar{\mathbf{u}})\,|e_{ij}| = \sigma_b(\bar{\mathbf{u}})\,|e_{kl}|\,, & l\in\bar{\mathcal{N}}_k^w\\ \mathbf{0}, & \text{otherwise}\end{cases} \tag{81}$$

Collecting for $k = 1,2,\ldots,N_B$ and $l = 1,2,\ldots,N_C$ leads to,

$$\bar{\mathbf{C}}\left(\bar{\mathbf{u}}\right) = \mathbf{W}_u\left(\boldsymbol{\Phi}^\dagger\mathbf{C}\left(\boldsymbol{\Phi}\bar{\mathbf{u}}\right)\boldsymbol{\Pi}\right) \in \mathbb{R}^{N_B\times N_C} \tag{82}$$

with $\mathbf{W}_u$ as a diagonal weight matrix with $\left[\mathbf{W}_u\right]_i = |\mathcal{V}_k|$. Coarse-graining the $\mathbf{E}\left(\mathbf{u}\right)$ matrix follows a similar procedure, where now $|\Omega_k^c| = |\mathcal{F}_k|\,|W_i|$, the neighbors on Ω_l^b of component k are $\bar{\mathcal{N}}_k^u = \{l|\mathcal{E}_{kl}^u \neq \emptyset\}$ with $\mathcal{E}_{kl}^u = \{(i,j)|i\in\mathcal{F}_k, j\in\mathcal{N}_i^u, j\in\mathcal{V}_l\}$, and therefore,

$$\begin{aligned}
\left[\boldsymbol{\Pi}^\dagger \mathbf{E}\left(\boldsymbol{\Phi}\bar{\mathbf{u}}\right)\boldsymbol{\Phi}\right]_{kl} &= \sum_{i\in\mathcal{F}_k}\sum_{j\in\mathcal{V}_l} \frac{|W_i|}{\left|\Omega_k^c\right|}\boldsymbol{\psi}_i^{k\top}\mathbf{E}_{ij}\left(\boldsymbol{\Phi}\bar{\mathbf{u}}\right)\varphi_j^l &\text{(83a)}\\
&= \sum_{i\in\mathcal{F}_k}\sum_{j\in\mathcal{V}_l} \frac{|W_i|}{\left|\Omega_k^c\right|}\left[\mathbf{E}_{ij}(\boldsymbol{\Phi}\bar{\mathbf{u}})\right]_{11} &\text{(83b)}\\
&= \frac{1}{|\mathcal{F}_k|}\sum_{i\in\mathcal{F}_k}\sum_{j\in\mathcal{V}_l}\left[\mathbf{B}_{ij}^j\right]_{11} &\text{(83c)}
\end{aligned}$$

From Eq. (81) it follows that,

$$\sum_{i\in\mathcal{F}_k}\sum_{j\in\mathcal{V}_l}\left[\mathbf{B}_{ij}^{j}\right]_{11}=\begin{cases}\sum_{p\in\bar{\mathcal{N}}_k^u}\sum_{(i,j)\in\mathcal{E}_{kp}^u}\sigma_b(\bar{\mathbf{u}})\,|e_{ij}|=\sigma_b(\bar{\mathbf{u}})\,|e_{kl}|\,, & l\in\bar{\mathcal{N}}_k^u\\ \mathbf{0}, & \text{otherwise}\end{cases} \tag{84}$$

Collecting for $k=1,2,\ldots,N_C$ and $l=1,2,\ldots,N_B$ leads to the following set of coarse-grained stiffness matrices,

$$\bar{\mathbf{B}}=\mathbf{W}_u\left(\mathbf{\Phi}^\dagger\mathbf{B}\left(\mathbf{\Phi}\bar{\mathbf{u}}\right)\mathbf{\Phi}\right)\in\mathbb{R}^{N_B\times N_B} \tag{85a}$$

$$\bar{\mathbf{C}}=\mathbf{W}_u\left(\mathbf{\Phi}^\dagger\mathbf{C}\left(\mathbf{\Phi}\bar{\mathbf{u}}\right)\mathbf{\Pi}\right)\in\mathbb{R}^{N_B\times N_C} \tag{85b}$$

$$\bar{\mathbf{E}}=\mathbf{W}_w\left(\mathbf{\Pi}^\dagger\mathbf{E}\left(\mathbf{\Phi}\bar{\mathbf{u}}\right)\mathbf{\Phi}\right)\in\mathbb{R}^{N_C\times N_B} \tag{85c}$$

$$\bar{\mathbf{F}}=\mathbf{W}_w\left(\mathbf{\Pi}^\dagger\mathbf{F}\left(\mathbf{\Pi}\bar{\mathbf{w}}\right)\mathbf{\Pi}\right)\in\mathbb{R}^{N_C\times N_C} \tag{85d}$$

Term #3: The heat-load vector $\boldsymbol{\mathcal{F}}(t)$ is coarse-grained next. To simplify the derivation, it is assumed that the heat load is applied only to the bulk sub-domains, i.e., $\boldsymbol{\mathcal{F}}(t)=\left[\mathbf{f}^{(b)}(t),\mathbf{0}\right]^\top$, given that the contact sub-domains are assumed to be thin and have negligible energy storage in subsequent developments. Therefore, the k-th element of the coarse-grained heat-load vector is computed as,

Coarse-Grained Heat-Load Vector

$$\begin{aligned}\left[\mathbf{\Phi}^\dagger\boldsymbol{\mathcal{F}}(t)\right]_k &= \sum_{i=1}^{B}\boldsymbol{\varphi}_i^{k\dagger}\mathbf{f}_i^{(b)}(t) && (86a)\\ &= \sum_{i\in\mathcal{V}_k}\frac{|B_i|}{|\Omega_k^b|}\boldsymbol{\varphi}_i^{k\top}\mathbf{f}_i^{(b)} && (86b)\\ &= \frac{1}{|\mathcal{V}_k|}\sum_{i\in\mathcal{V}_k}\left[\mathbf{f}_i^{(b)}\right]_1 && (86c)\\ &= \frac{1}{|\mathcal{V}_k|}\sum_{i\in\mathcal{V}_k}|e_{iq}|\,\bar{q}_{i,b}(t) && (86d)\\ &= \frac{1}{|\mathcal{V}_k|}\,|e_{kq}|\,\bar{q}_{k,b}(t) && (86e)\end{aligned}$$

Collecting for $k=1,2,\ldots,N_B$ leads to the following coarse-grained heat-load vector,

$$\bar{\mathbf{F}}(t)=\mathbf{W}_u\left(\mathbf{\Phi}^\dagger\boldsymbol{\mathcal{F}}(t)\right)\in\mathbb{R}^{N_B} \tag{87}$$

By collecting the coarse-grained capacitance and stiffness matrices, and applying the zero-energy storage assumption, the following DAE system is obtained,

$$\bar{\mathbf{A}}\left(\bar{\mathbf{u}}\right)\dot{\bar{\mathbf{u}}}=\bar{\mathbf{B}}\bar{\mathbf{u}}+\bar{\mathbf{C}}\bar{\mathbf{w}}+\bar{\mathbf{F}}(t) \tag{88a}$$

$$\mathbf{0}=\bar{\mathbf{E}}\left(\bar{\mathbf{u}}\right)\bar{\mathbf{u}}+\bar{\mathbf{F}}\bar{\mathbf{w}} \tag{88b}$$

where Eq. (88b) may be re-arranged to match precisely the energy conservation from Eq. (15). Specifically, the k-th equation of Eq. (88b) states that,

$$0 = \sum_{p \in \bar{\mathcal{N}}_k^u} \sigma_b(\bar{\mathbf{u}}) \, |e_{kp}| \, (\bar{u}_p - \bar{w}_k) - \sum_{p \in \bar{\mathcal{N}}_k^c} \sigma_c \, |\Gamma_{kp}| \, (\bar{w}_k - \bar{w}_p) \tag{89}$$

Now, note that each domain is decomposed as $\Omega_k = \Omega_k^b \cup \Omega_k^c$, and that Γ_r for $r = 1, 2, \ldots, N_\Gamma$ is composed of two unique interface domains $\Omega_r^{c,-} \cup \Omega_r^{c,+}$. Therefore, Eq. (89) simplifies to,

$$0 = \sigma_b(\bar{\mathbf{u}}) \, |e_{kk}| \, (\bar{u}_k - \bar{w}_k) - \sigma_{c,r} \, |\Gamma_r| \, \Delta \bar{w}_r \tag{90}$$

which corresponds precisely to the heat flux continuity constraint in Eq. (15), from which Eq. (16) is derived. The derivation of Eq. (22) from the coarse-grained DG-FEM model is further demonstrated next in the following example.

Coarse-Graining Example Consider the three-layered TPS shown in Fig. [X] with three components $\{\Omega_j\}_{j=1}^3$ and three contact interfaces $\{\Gamma_k\}_{k=1}^3$. The set of elements for each component, and on the minus and plus sides of each interface are defined as,

$$\mathcal{V}_j = \{m \in \{1, \ldots, M\} : E_m \subseteq \Omega_j\}, \tag{91a}$$

$$\mathcal{F}_k^- = \left\{m \in \{1, \ldots, M\} : E_m \cap \Gamma_k \neq \emptyset, E_m \subseteq \Omega_{l(k)}\right\}, \tag{91b}$$

$$\mathcal{F}_k^+ = \left\{m \in \{1, \ldots, M\} : E_m \cap \Gamma_k \neq \emptyset, E_m \subseteq \Omega_{r(k)}\right\} \tag{91c}$$

where $\Omega_{l(k)}$ and $\Omega_{r(k)}$ are the left and right components that form the interface Γ_k, respectively, with $\Gamma_k = \partial\Omega_{l(k)} \cap \partial\Omega_{r(k)}$.

Important point: The critical assumption is that the contact domains $\Omega_r^{c,-}$ and $\Omega_r^{c,+}$ are sufficiently thin such that their transient thermal response is negligible, and thus the contact temperature jumps $\Delta\bar{\theta}_r$ are governed by algebraic constraints for heat-flux continuity across the interfaces.